HIKITA'S $\star$-PRODUCT ON SYMMETRIC FUNCTIONS: PIERI RULES, DOMINANCE SUPPORT, AND THE $(s-1)$-ADIC VALUATION

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Abstract. Hikita introduced a two-parameter deformation $\star$ of the product of symmetric functions, obtained by transporting the ordinary product through the level-one polynomial representation of the affine Hecke algebra. We give an explicit rule for $e_k \star (e_{a_1} \cdots e_{a_\ell})$ for all k and ℓ, in generating-function form; its case $\ell = 1$ is a Pieri rule for $e_k \star e_r$ generalising Hikita's rule for $e_1 \star e_r$. Writing $e_{\lambda_1} \star \cdots \star e_{\lambda_\ell} = \sum_\mu c_{\lambda\mu}(s,t) e_\mu$, we show that $c_{\lambda\mu}$ is supported exactly on the dominance up-set of λ with s-adic valuation $n(\mu)$, and that the $(s-1)$-adic valuation of $c_{\lambda\mu}$ is $\ell(\lambda) - \kappa(\lambda,\mu)$, where κ is the maximal number of blocks into which λ and μ split compatibly. Leading coefficients factor over blocks into connected pieces; for a full merge the connected piece is a connected-graph cumulant. The coefficients are invariant, up to a power of s, under complementation in a box, and all second-order leading coefficients are given in closed form via a residue formula for Green polynomials at two-part classes.

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1. Introduction

Hikita [Hik25] defined a commutative, associative product $\star$ on symmetric functions over $\mathbb{Q}(q,t)$ by transporting the ordinary product through the level-one polynomial representation of the affine Hecke algebra. We write $s := q^{-1}$. At $s = 1$, $\star$ is the ordinary product. Hikita's Pieri rule [Hik25, Thm. 3.12] reads, in our normalisation,

$$e_1 \star e_r = (1-s)[r+1]_t \, e_{r+1} + s \, e_1 e_r.$$

Di Francesco and Kedem [DFK19, §8.3] observed that the associated t-deformed graded characters lose Schur positivity at generic t. Chen, Lu and Ruan [CLR26] identify the derived Hall

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algebra of the Jordan quiver with a ring of double symmetric functions with parameter t, sending the derived Hall basis to double Hall–Littlewood functions, for which they also prove Pieri rules; this is a related Hall-algebra product, and we do not compare it with $\star$ here. This paper is about the structure that survives: explicit multiplication rules, triangularity, and valuations.

Pieri rules. The operator $e_k \star (\cdot)$ is a sum over k-subsets (Theorem 2.5),

$$e_k \star F = \sum_{|A|=k} \prod_{i\in A,\, j\notin A} \frac{x_i - t x_j}{x_i - x_j} \prod_{i\in A} x_i \;\cdot F\big|_{x_i \mapsto s x_i \;(i\in A)}.$$

From it we derive a closed generating function for $e_k \star (e_{a_1} \cdots e_{a_\ell})$ for all k, ℓ (Theorem 3.2). Since the e_a generate, this determines $e_k \star F$ for every F. The case $\ell = 1$ is a Pieri rule (Corollary 3.3):

$$e_k \star e_r = \sum_{b=0}^{k} s^b \, F_{k-b}(t^{r-b}) \, e_b \, e_{r+k-b},$$

with explicit polynomials $F_n(w)$, and at $t = 0$ the coefficients form a truncated geometric distribution (Corollary 3.6). In the generating function, the ℓ columns interact only through a pairwise cross kernel, and the proof closes with a single residue identity (Lemma 3.9).

Dominance and valuations. Write $e^\star_\lambda := e_{\lambda_1} \star \cdots \star e_{\lambda_\ell} = \sum_\mu c_{\lambda\mu}(s,t)\, e_\mu$. A termwise degree count on the subset formula shows that $c_{\lambda\mu} \in \mathbb{Q}[s,t]$ is supported exactly on the dominance upset of λ, with s-adic valuation exactly $n(\mu)$ (Theorem 4.1). In the rescaled basis $b_\mu = s^{n(\mu)} e_\mu$, the operator $e_k \star (\cdot)$ reduces modulo s to Hall–Littlewood multiplication by $t^{-\binom{k}{2}} e_k$, so the s-leading coefficients form the classical $e \to$Hall–Littlewood transition matrix (Theorem 5.1 and Corollary 5.2).

At $s = 1$, $\star$ is the ordinary product, and the variable s enters the subset formula only through an Euler operator; so the $(s-1)^p$ Taylor piece of $e_k \star (\cdot)$ is a differential operator of order p. The first-order term is a symmetric biderivation with an explicit matrix (Theorem 6.1 and Proposition 6.2), and an order-p piece can glue at most p existing blocks of parts. This gives the *block valuation law*

$$v_{s-1}(c_{\lambda\mu}) = \ell(\lambda) - \kappa(\lambda,\mu) \qquad (\mu \trianglerighteq \lambda,\ \mu \neq \lambda),$$

where κ is the maximal number of compatible blocks (Theorem 6.6); the lower bound holds for all t, and exactness is settled at $t = 0$, where no cancellation occurs. The leading coefficients are sums over merge histories with explicit weights (Theorems 6.9 and 6.10). For a full merge the lead is $\frac{1-t^n}{\prod_i (1-t^{\lambda_i})} \sum_{H \text{ connected}} \prod_{ij\in H} (t^{\lambda_i\lambda_j} - 1)$ (Theorem 7.4), whose graph half is due to Dołęga [Doł19], and every lead is a sum of products of connected leads (Theorems 7.7 and 7.8). The coefficients satisfy $c_{N^\ell - \lambda, N^\ell - \mu} = s^{N\binom{\ell}{2} - (\ell-1)|\lambda|} c_{\lambda\mu}$ (Theorem 8.1). Finally, every leading coefficient at valuation 2 is given in closed form (Theorem 9.7 and Corollary 9.9), via a constant-term formula and a residue evaluation of Green polynomials at two-part classes (Theorems 9.2 and 9.3).

Organisation. Section 2 sets up Hikita's product and proves the subset formula. Section 3 contains the Pieri package. Section 4 proves dominance support, and Section 5 identifies the $s \to 0$ edge. Sections 6 and 7 treat the expansion at $s = 1$, Section 8 the box symmetry and the linear coefficient, and Section 9 the second-order leading coefficients.

2. Hikita's product and the subset formula

2.1. **Notation.** For $m \geq 0$ let $\Lambda_m := \mathbb{Q}(s,t)[x_1,\dots,x_m]^{S_m}$ and let Λ be the ring of symmetric functions over $\mathbb{Q}(s,t)$. We write e_r for the elementary symmetric functions ($e_0 = 1$, $e_r = 0$ for $r < 0$), $e_\lambda = e_{\lambda_1} e_{\lambda_2} \cdots$, and

$$E(z) := \prod_{i=1}^{m} (1 + x_i z) = \sum_{r \geq 0} e_r z^r, \qquad Q(y) := \frac{E(-ty)}{E(-y)} = \prod_{i=1}^{m} \frac{1 - t x_i y}{1 - x_i y} = \sum_{n \geq 0} q_n y^n,$$

so that the q_n are the one-row Hall–Littlewood functions of [Mac95, III (2.10)]. For a set B of indices, $E(x_B; z) := \prod_{i \in B}(1 + x_i z)$, and $\hat{x}_i$ denotes the variables $x_1, \dots, x_m$ with x_i omitted. We put

$$a_{ij} := \frac{x_i - t x_j}{x_i - x_j}, \qquad c_A := \prod_{i \in A,\ j \notin A} a_{ij}, \qquad x_A := \prod_{i \in A} x_i \qquad (A \subseteq [m]).$$

Further, $[n]_q = (1-q^n)/(1-q)$ and $[n] = [n]_t$, $[n]_t! = [1][2]\cdots[n]$, $(a;t)_n = \prod_{i=0}^{n-1}(1 - a t^i)$, $\varphi_r(t) = \prod_{a=1}^{r}(1 - t^a)$, $n(\lambda) = \sum_i (i-1)\lambda_i$, $m_v(\lambda) = \#\{i : \lambda_i = v\}$, and $\trianglerighteq$ is dominance order. $P_\lambda(x;t)$ and $Q_\lambda = b_\lambda(t) P_\lambda$, $b_\lambda(t) = \prod_{v \geq 1} \varphi_{m_v(\lambda)}(t)$, are the Hall–Littlewood functions of [Mac95, III §2]; $K_{\lambda\mu}$ are the Kostka numbers and $K_{\lambda\mu}(t)$ the Kostka–Foulkes polynomials, $s_\lambda = \sum_\mu K_{\lambda\mu}(t) P_\mu(x;t)$ [Mac95, III (2.6)]. We use two scalar products on Λ: the Hall pairing $\langle p_\lambda, p_\mu \rangle = \delta_{\lambda\mu} z_\lambda$ [Mac95, I §4] and the Hall–Littlewood pairing $\langle p_\lambda, p_\mu \rangle_t = \delta_{\lambda\mu} z_\lambda \prod_i (1 - t^{\lambda_i})^{-1}$ [Mac95, III §4]. They are different: an undecorated bracket $\langle \cdot, \cdot \rangle$ is always the Hall pairing.

2.2. **The level-one representation and Hikita's product.** The symmetric group S_m acts on $\mathbb{Q}(s,t)[x_1,\dots,x_m]$ by permuting variables; s_i is the transposition of x_i and x_{i+1}. Define operators on $\mathbb{Q}(s,t)(x_1,\dots,x_m)$ by

$$\begin{aligned} T_i f &:= t\, s_i f + (t-1)\, \frac{x_{i+1}}{x_i - x_{i+1}} (s_i f - f) \qquad (1 \leq i \leq m-1), \\ \pi f(x_1,\dots,x_m) &:= x_1\, f(x_2,\dots,x_m, s x_1), \\ Y_i &:= t^{m-i}\, T_{i-1} \cdots T_1\, \pi\, T_{m-1}^{-1} \cdots T_i^{-1} \qquad (1 \leq i \leq m). \end{aligned}$$

The T_i are Demazure–Lusztig operators; they preserve polynomials, satisfy $(T_i - t)(T_i + 1) = 0$ and the braid relations, and $T_i f = t f$ whenever $s_i f = f$. For $1 \leq k \leq m$ put

$$e_k(Y) := \sum_{b_1 < \cdots < b_k} Y_{b_1} Y_{b_2} \cdots Y_{b_k},$$

each product taken in increasing order of indices from left to right. (The Y_i commute, but the proof of Theorem 2.5 does not use this.) Hikita's product $F \star G$ is defined by transporting the product through the map $F(Y) \mapsto F(Y) \bullet 1$; by [Hik25, Lemma 3.1, Cor. 3.9] this map is a linear isomorphism compatible with $m \to m-1$, so $\star$ is well defined on Λ [Hik25, Cor. 3.10], and it is commutative and associative [Hik25, §3.2, after Def. 3.4]. We use $\star$ only through the following consequence of the definitions [Hik25, Def. 3.4, Lemma 3.3]: for symmetric F,

$$e_k \star F = t^{-\binom{k}{2}}\, e_k(Y)\, F. \tag{2.1}$$

2.3. **Coxeter preliminaries.** We use standard facts about S_m as a Coxeter group with generators $s_1, \dots, s_{m-1}$ [Hum90, §§1.10, 5.12], [BB05, §2.4]. Words act as compositions of operators, the rightmost factor first.

(H1) For a reduced word $w = s_{i_1} \cdots s_{i_\ell}$ the operator $T_w := T_{i_1} \cdots T_{i_\ell}$ depends only on w, and $T_u T_v = T_{uv}$ whenever $\ell(uv) = \ell(u) + \ell(v)$.

(H2) For $J \subseteq \{s_1, \dots, s_{m-1}\}$ let W_J be the parabolic subgroup and $W^J = \{w : \ell(ws) > \ell(w)\ \forall s \in J\}$. Every w factors uniquely as $w = w^J w_J$ with $w^J \in W^J$, $w_J \in W_J$, and $\ell(w) = \ell(w^J) + \ell(w_J)$.
(H3) For $K \subseteq J$, multiplication $W^J \times (W_J)^K \to W^K$ is a length-additive bijection.
(H4) If $T_j H = tH$ for all $s_j \in J$, then $T_v H = t^{\ell(v)} H$ for all $v \in W_J$.
(H5) $\pi T_j = T_{j+1}\pi$ for $1 \le j \le m-2$.

(H3) is standard: if $u \in W^J$, $v \in (W_J)^K$ and $s \in K$, then $\ell(uvs) = \ell(u) + \ell(vs) = \ell(uv) + 1$; conversely, for $x \in W^K$ write $x = x^J x_J$, and $x_J s < x_J$ for some $s \in K$ would give $\ell(xs) < \ell(x)$. (H4) is immediate from (H1). For (H5), $\pi s_j f = x_1 f(x_2, \dots, x_{j+2}, x_{j+1}, \dots, sx_1) = s_{j+1}\pi f$ because $j+2 \le m$, and π turns multiplication by $x_{j+1}/(x_j - x_{j+1})$ into multiplication by $x_{j+2}/(x_{j+1} - x_{j+2})$.

Let $J_n := \{s_i : i \ne n\}$, so $W_{J_n} = S_n \times S_{m-n}$, and the coset wW_{J_n} is determined by $D = w([n])$. Its minimal element w^D is increasing on $[n]$ and on $[n+1, m]$, with $\ell(w^D) = \sum_l (d_l - l)$ for $D = \{d_1 < \cdots < d_n\}$. The word

$$w(D) := (s_{d_1 - 1} \cdots s_1)(s_{d_2 - 1} \cdots s_2) \cdots (s_{d_n - 1} \cdots s_n)$$

sends $l \mapsto d_l$ for $l \le n$: the blocks with index $l' > l$ fix l, block l sends l to d_l, and the blocks $l' < l$ only move points $\le d_{l'} < d_l$. It has length $\le \sum_l (d_l - l)$, so $w(D) = w^D$ and the word is reduced. We write $T_{w(D)}$ for the corresponding operator and

$$\sigma^{(k)} := \sum_{|D| = k} T_{w(D)}.$$

2.4. **The parabolic kernel.**

Lemma 2.1. *Let $0 \le n \le m$, $1 \le c < d_1 < \cdots < d_n \le m$, and let H satisfy $T_j H = tH$ for all $j \ne n$. Then*

$$(T_c T_{c+1} \cdots T_{m-1})^{-1} T_{w(D)} H = t^{\,n-(m-c)}\, T_{w(D-1)} H, \qquad D - 1 := \{d_1 - 1, \dots, d_n - 1\}.$$

Proof. Let $x := s_c s_{c+1} \cdots s_{m-1}$; it is reduced of length $m - c$, sends $m \mapsto c$ and $i \mapsto i + 1$ for $c \le i \le m-1$, and fixes $i < c$. Let $y := w(D-1)$; the word is valid since $d_l - 1 \ge c + l - 1 \ge l$, and $\ell(y) = \ell(w^D) - n$. As $m \notin D - 1$, the value m is the largest on the increasing block $[n+1, m]$, so $y(m) = m$. Since x is increasing on $[1, m-1]$, the inversions of y among positions $< m$ survive in xy; at position m, $xy(m) = c$ is inverted with exactly those $i < m$ having $y(i) \ge c$, and there are $m - c$ of them. Hence $\ell(xy) = \ell(y) + m - c$. Finally $xy([n]) = x(D-1) = D$ because every $d_l - 1 \ge c$, so $xy = w^D v$ with $v \in W_{J_n}$ and $\ell(v) = \ell(xy) - \ell(w^D) = m - c - n$. By (H1) and (H4), $T_x T_y H = T_{xy} H = T_{w^D} T_v H = t^{m-c-n} T_{w^D} H$. Apply T_x^{-1}. □

Proposition 2.2 (Parabolic kernel)**.** *For symmetric F and $1 \le b_1 < \cdots < b_k \le m$, $Y_{b_1} Y_{b_2} \cdots Y_{b_k} F = t^{\binom{k}{2}}\, T_{w(\{b_1, \dots, b_k\})}\, \pi^k F$. Hence*

$$e_k(Y) F = t^{\binom{k}{2}}\, \sigma^{(k)} \pi^k F.$$

Proof. Note that $\pi^n F = x_1 \cdots x_n\, F(x_{n+1}, \dots, x_m, sx_1, \dots, sx_n)$ is symmetric in $x_1, \dots, x_n$ and in $x_{n+1}, \dots, x_m$, so $T_j \pi^n F = t \pi^n F$ for $j \ne n$. We induct on the number of factors, the empty product being trivial. Let $c < d_1 < \cdots < d_n$ and assume $Y_{d_1} \cdots Y_{d_n} F = t^{\binom{n}{2}} T_{w(D)} H$ with $H = \pi^n F$. By Lemma 2.1,

$$Y_c\, t^{\binom{n}{2}} T_{w(D)} H = t^{\binom{n}{2}} t^{m-c}\, T_{c-1} \cdots T_1\, \pi\, (T_c \cdots T_{m-1})^{-1} T_{w(D)} H = t^{\binom{n}{2} + n}\, T_{c-1} \cdots T_1\, \pi\, T_{w(D-1)} H.$$

All indices in the word $w(D-1) = \prod_l (s_{d_l - 2} \cdots s_l)$ lie in $[1, m-2]$, so by (H5), $\pi T_{w(D-1)} = \prod_l (T_{d_l - 1} \cdots T_{l+1})\, \pi$. Now $T_{c-1} \cdots T_1 \prod_l (T_{d_l - 1} \cdots T_{l+1}) = T_{w(\{c\} \cup D)}$, $\pi H = \pi^{n+1} F$, and $\binom{n}{2} + n = \binom{n+1}{2}$. □

2.5. **The k-set kernel.** Let $\sigma_{[l,m]} := \sum_{a=l}^{m} T_{a-1}\cdots T_l$ be the partial symmetriser on $x_l,\dots,x_m$.

Lemma 2.3. *Let F be symmetric in $x_2,\dots,x_m$, with coefficients in any field containing $\mathbb{Q}(t)$, and let $F^{(i)}$ be F with x_1 and x_i swapped. Then $\sigma_{[1,m]}F = \sum_{i=1}^{m} F^{(i)} \prod_{j\neq i} a_{ij}$.*

Proof. Induction on m; $m=1$ is trivial. Since $s_1F = F^{(2)}$, a direct computation gives $T_1F = a_{21}F^{(2)} + (a_{12}-1)F^{(1)}$. Write $\sigma_{[1,m]} = 1 + \sigma_{[2,m]}T_1$. The function $G := T_1F$ is symmetric in $x_3,\dots,x_m$, so by induction (over the field with x_1 adjoined) $\sigma_{[2,m]}G = \sum_{i\geq 2} G|_{x_2\leftrightarrow x_i} \prod_{j\geq 2,\, j\neq i} a_{ij}$. Under $x_2\leftrightarrow x_i$, $F^{(2)}\mapsto F^{(i)}$, $F^{(1)}$ is fixed, $a_{21}\mapsto a_{i1}$ and $a_{12}\mapsto a_{1i}$. Hence

$$\sigma_{[1,m]}F = \sum_{i\geq 2} F^{(i)} \prod_{j\neq i} a_{ij} + F^{(1)}\Big[1 + \sum_{i\geq 2}(a_{1i}-1) \prod_{j\geq 2,\, j\neq i} a_{ij}\Big].$$

The function $R(x) := \prod_{j\geq 2}(x - tx_j)/(x-x_j)$ tends to 1 at ∞ and has simple poles at $x=x_i$ with residues $(1-t)x_i\prod_{j\geq 2, j\neq i} a_{ij}$; so $R(x) = 1 + \sum_{i\geq 2} \frac{(1-t)x_i}{x-x_i} \prod_{j\geq 2, j\neq i} a_{ij}$. Evaluating at $x=x_1$ and using $(1-t)x_i/(x_1-x_i) = a_{1i}-1$ shows that the bracket equals $R(x_1) = \prod_{j\geq 2} a_{1j}$. □

Proposition 2.4 (k-set kernel)**.** *Let G be symmetric in $x_1,\dots,x_k$ and in $x_{k+1},\dots,x_m$, and let $G^{(A)}$ denote G with the variables x_A in the first k slots and $x_{[m]\setminus A}$ in the last $m-k$. Then*

$$\sigma^{(k)}G = \sum_{|A|=k} c_A\, G^{(A)}.$$

Proof. Let $K_l := \{s_{l+1},\dots,s_{m-1}\}$. The minimal coset representatives of $W_{K_{l-1}}/W_{K_l}$ are $s_{a-1}\cdots s_l$ $(l\leq a\leq m)$, so applying (H3) along $K_k\subset K_{k-1}\subset\cdots\subset K_0$ gives $\sum_{u\in W^{K_k}} T_u = \sigma_{[1,m]}\sigma_{[2,m]}\cdots\sigma_{[k,m]}$. Applying (H3) to $K_k\subset J_k$, with $(W_{J_k})^{K_k} = S_k$, gives $\sum_{u\in W^{K_k}} T_u = \sigma^{(k)}\sum_{v\in S_k} T_v$. For G as in the statement, (H4) gives $\sum_{v\in S_k} T_vG = [k]_t!\, G$, so

$$\sigma_{[1,m]}\cdots\sigma_{[k,m]}G = [k]_t!\,\sigma^{(k)}G. \tag{2.2}$$

On the other hand, if G is merely symmetric in $x_{k+1},\dots,x_m$, then iterating Lemma 2.3 gives

$$\sigma_{[1,m]}\cdots\sigma_{[k,m]}G = \sum_{(a_1,\dots,a_k)\ \text{distinct}} G(x_{a_1},\dots,x_{a_k};\text{rest})\ \omega(a), \qquad \omega(a) := \prod_{l=1}^{k}\ \prod_{j\notin\{a_1,\dots,a_l\}} a_{a_l j}. \tag{2.3}$$

Indeed, by induction on k in the variables $x_2,\dots,x_m$, $H := \sigma_{[2,m]}\cdots\sigma_{[k,m]}G$ is the analogous sum over distinct tuples in $[2,m]$; it is symmetric in $x_2,\dots,x_m$, since permuting those variables permutes the terms; and Lemma 2.3 applied to H, followed by $x_1\leftrightarrow x_{a_1}$, turns tuples in $[2,m]$ into tuples in $[m]\setminus\{a_1\}$.

Now let G be symmetric in both groups. Then $G(x_{a_1},\dots,x_{a_k};\text{rest}) = G^{(A)}$ with $A = \{a_1,\dots,a_k\}$, and

$$\omega(a) = c_A \prod_{1\leq l<l'\leq k} a_{a_l a_{l'}}. \tag{2.4}$$

By the Poincaré identity $\sum_{w\in S_k} w\big(\prod_{i<j\leq k} a_{ij}\big) = [k]_t!$ [Mac95, III (1.4)] (which is also (2.2)=(2.3) at $m=k$, $G=1$), summing over the orderings of A gives $[k]_t!\, c_A G^{(A)}$. Compare with (2.2). □

Theorem 2.5 (Subset formula)**.** *For every $m\geq 0$, $k\geq 1$ and $F\in\Lambda_m$,*

$$e_k \star F = t^{-\binom{k}{2}} e_k(Y)F = E_kF := \sum_{|A|=k} c_A\, x_A\, F\big|_{x_i\mapsto sx_i\ (i\in A)}.$$

Proof. By Propositions 2.2 and 2.4 with $G = \pi^k F$, since $(\pi^k F)^{(A)} = x_A\, F|_{x_i\mapsto sx_i\,(i\in A)}$; the first equality is (2.1). For $m<k$ both sides vanish. □

Remark 2.6. For $k=1$ the subset formula gives back Hikita's rule $e_1\star e_r=(1-s)[r+1]e_{r+1}+se_1e_r$ [Hik25, Thm. 3.12]; see Example 3.4. Proposition 2.4 is the level-one form of the classical expression of the Hecke symmetriser as $\sum_w w\circ\prod_{i<j}a_{ij}$, restricted to minimal coset representatives.

Lemma 2.7. $E_k(1)=\sum_{|A|=k}c_Ax_A=e_k$.

Proof. Grouping $w\in S_m$ by $w(\{1,\dots,k\})=A$ and applying the Poincaré identity to $S_k\times S_{m-k}$ shows that $\sum_{|A|=k}c_Ax_A$ is the Hall–Littlewood function $P_{(1^k)}(x;t)$ as defined in [Mac95, III (2.1)–(2.2)], and $P_{(1^k)}=e_k$ [Mac95, III (2.8)]. □

Lemma 2.8 (Stability)**.** *For $F\in\Lambda_m$, $(E_kF)|_{x_m=0}=E_k(F|_{x_m=0})$, the right side computed in $m-1$ variables. Hence the e-coefficients of any expression built from the E_k and 1 are independent of m once m is at least the degree.*

Proof. Terms with $m\in A$ vanish because of the factor x_A; for $m\notin A$, $a_{im}|_{x_m=0}=1$. Restriction $\Lambda_m\to\Lambda_{m-1}$ maps $e_\nu\mapsto e_\nu$ and is injective in degrees $\le m-1$. □

2.6. **Two consequences of the kernel.** The next two lemmas drive every computation in Section 3. For a function φ of one variable put $S_k^{\varphi}(x_1,\dots,x_m):=\sum_{|A|=k}c_A\prod_{a\in A}\varphi(x_a)$.

Lemma 2.9 (Peeling one variable)**.** *For $m\ge1$ and $k\ge1$,*

$$[k]\,S_k^{\varphi}(x)=\sum_{i=1}^{m}\varphi(x_i)\prod_{j\neq i}a_{ij}\;S_{k-1}^{\varphi}(\hat{x}_i).$$

Proof. By (2.4) and the Poincaré identity, $[k]_t!\,S_k^{\varphi}=\sum_a\omega(a)\prod_l\varphi(x_{a_l})$, the sum over distinct k-tuples. Separate $a_1=i$: $\omega(a)=\prod_{j\neq i}a_{ij}\cdot\omega'(a_2,\dots,a_k)$, where ω' is the weight of (2.3) for the variables $\hat{x}_i$. Summing over $(a_2,\dots,a_k)$ gives $[k-1]_t!\,S_{k-1}^{\varphi}(\hat{x}_i)$. □

Lemma 2.10 (Residue form of the HL kernel)**.** *Let $\mathbb{K}\supseteq\mathbb{Q}(t)$ be a field and $H\in\mathbb{K}(v)$ with $H(0)=0$, such that $H(v)/v$ has only simple poles, at points $p\in\mathbb{K}$, and $H(v)/v=O(1/v)$ at ∞. Then in $\mathbb{K}(x_1,\dots,x_m)$*

$$(1-t)\sum_{i=1}^{m}H(x_i)\prod_{j\neq i}a_{ij}=-\sum_{p}\operatorname*{Res}_{v=p}f-\operatorname*{Res}_{v=\infty}f,\qquad f(v):=\frac{H(v)}{v}\prod_{j=1}^{m}\frac{v-tx_j}{v-x_j}.$$

Proof. f is regular at $v=0$, its residue at $v=x_i$ is $(1-t)H(x_i)\prod_{j\neq i}a_{ij}$, and the sum of all residues of a rational function is zero. □

For $H(v)=v^n$, $n\ge1$, this is $(1-t)\sum_i x_i^n\prod_{j\neq i}a_{ij}=q_n$ [Mac95, III (2.10)]. Note that $\prod_j(v-tx_j)/(v-x_j)=Q(1/v)$ and $Q(-a)=E(ta)/E(a)$.

3. The Pieri package

3.1. **The coefficients.** For $j\ge0$ put

$$\alpha_j:=\prod_{i=1}^{j}\frac{s-t^i}{1-t^i}\quad(\alpha_{-1}:=0),\qquad c(n,j):=\frac{(s;t)_{n-j}}{(t;t)_{n-j}}(\alpha_j-s\,t^{n-j}\alpha_{j-1})\quad(0\le j\le n),$$

and $c(n,j):=0$ otherwise. Let $F_n(w):=\sum_{j=0}^{n}c(n,j)w^j$. For instance $F_0=1$ and $F_1(w)=(1-s)(1-tw)/(1-t)$.

Lemma 3.1. *Let $G(x):=\sum_{n\ge0}\frac{(s;t)_n}{(t;t)_n}x^n$, $C_j(x):=\sum_n c(n,j)x^n$, and $D_j:=\alpha_j-s\alpha_{j-1}$. Then*

$$(1-x)G(x)=(1-sx)G(tx),\qquad C_j(x)=D_j\,x^j\,G(tx)\,\frac{1-st^{-j}x}{1-x},$$

$D_0=1$, $D_j=\alpha_{j-1}t^j(s-1)/(1-t^j)\neq0$ for $j\ge1$, and $(1-t^j)D_j=(ts-t^j)D_{j-1}$ for $j\ge1$.

Proof. Write $g_n = (s;t)_n/(t;t)_n$. The first identity says $g_n - g_{n-1} = t^n g_n - st^{n-1}g_{n-1}$, i.e. $(1-t^n)g_n = (1-st^{n-1})g_{n-1}$. By definition $C_j(x) = x^j(\alpha_j G(x) - s\alpha_{j-1}G(tx)) = x^j G(tx)(\alpha_j(1-sx) - s\alpha_{j-1}(1-x))/(1-x)$. The bracket is linear in x. It equals D_j at $x=0$, and at $x = t^j/s$ it equals $\alpha_j(1-t^j) - \alpha_{j-1}(s-t^j) = 0$ (for $j=0$ use $\alpha_{-1}=0$). So it equals $D_j(1-st^{-j}x)$. The formula for D_j follows from $\alpha_j = \alpha_{j-1}(s-t^j)/(1-t^j)$, and the recursion from it and the same relation one step down (for $j=1$ use $D_0 = 1$). □

3.2. **Statements.** Fix $\ell \geq 1$ and indeterminates $z_1, \dots, z_\ell$. For integers $i, j \geq 0$ let

$$K_{ij}(z,w) := \prod_{p=0}^{i-1} \frac{t^p z - sw}{t^p z - t^j w} \prod_{r=0}^{j-1} \frac{sz - t^r w}{t^i z - t^r w},$$

and note $K_{ij}(z,w) = K_{ji}(w,z)$. For $I = (i_1, \dots, i_\ell) \in \mathbb{Z}^\ell_{\geq 0}$ with $|I| = S$ put $K_I := \prod_{c<c'} K_{i_c i_{c'}}(z_c, z_{c'})$ and

$$V_I^{(n)} := K_I \, s^{(\ell-1)(n-S)} \sum_{n_1+\cdots+n_\ell = n} t^{-\sum_{c\neq c'}(n_c - i_c)i_{c'}} \prod_{c=1}^{\ell} t^{-n_c i_c} \, c(n_c, i_c) \, z_c^{-n_c},$$

with $V_I^{(n)} := 0$ if some $i_c < 0$. Since $c(n_c, i_c) = 0$ for $n_c < i_c$, $V_I^{(n)} = 0$ unless $S \leq n$.

Theorem 3.2 (ℓ-column rule)**.** *For all $m, k \geq 0$ and $\ell \geq 1$, in $\Lambda_m \otimes \mathbb{Q}(s,t)(z_1, \dots, z_\ell)$,*

$$\mathcal{E}_k := \sum_{a \in \mathbb{Z}^\ell_{\geq 0}} z_1^{a_1} \cdots z_\ell^{a_\ell} \; e_k \star (e_{a_1} \cdots e_{a_\ell}) = \sum_{b=0}^{k} \sum_{I} \left(s^\ell t^{-|I|}\right)^b V_I^{(k-b)} \; e_b \prod_{c=1}^{\ell} E(t^{i_c} z_c).$$

In particular the right-hand side is a polynomial in $z_1, \dots, z_\ell$, and $e_k \star (e_{a_1} \cdots e_{a_\ell})$ is its coefficient of z^a. The coefficients do not depend on m, so the identity holds in Λ.

Corollary 3.3 (Pieri rule for $e_k \star e_r$)**.** *For all $k \geq 1$ and $r \geq 0$,*

$$e_k \star e_r = \sum_{b=0}^{k} s^b \, F_{k-b}(t^{r-b}) \, e_b \, e_{r+k-b}.$$

Equivalently, $\sum_r z^r \, e_k \star e_r = \sum_{j+b \leq k} s^b t^{-kj} c(k-b, j) \, e_b \, z^{b-k} E(t^j z)$.

Proof. Take $\ell = 1$ in Theorem 3.2: $K_I = 1$, and $V^{(n)}_{(j)} = t^{-nj} c(n,j) z^{-n}$, so the right-hand side is the stated generating function. Extract $[z^r]$ using $[z^r] z^{b-k} E(t^j z) = t^{j(r+k-b)} e_{r+k-b}$. □

For $r < k$, distinct b may give the same monomial $e_b e_{r+k-b}$; the identity is between the sums. Since $\star$ is commutative, applying Corollary 3.3 to $e_r \star e_k$ shows that the support of $e_k \star e_r$ is contained in $\{e_b e_{r+k-b} : 0 \leq b \leq \min(k, r)\}$ (see also Theorem 4.1).

Example 3.4 ($k = 1$)**.** $c(1,0) = \frac{1-s}{1-t}$ and $c(1,1) = \frac{t(s-1)}{1-t}$, so $F_1(t^r) = (1-s)[r+1]$ and Corollary 3.3 is Hikita's rule $e_1 \star e_r = (1-s)[r+1]e_{r+1} + s e_1 e_r$ [Hik25, Thm. 3.12].

Example 3.5 ($k = 2$)**.** One computes $F_2(w) = \dfrac{(1-s)(1-t^2w)\big(1-st+(s-t)w\big)}{(1-t)(1-t^2)}$, so for all $r \geq 0$

$$e_2 \star e_r = \frac{(1-s)(1-t^{r+2})(1-st+(s-t)t^r)}{(1-t)(1-t^2)} \, e_{r+2} + s(1-s)[r] \, e_1 e_{r+1} + s^2 e_2 e_r.$$

For $r = 0$ this is $e_2 \star 1 = e_2$.

Corollary 3.6 (The case $t=0$)**.** *For $k\ge 1$, $r\ge 0$, put $\mu=\min(k,r)$ and $M=\max(k,r)$. The coefficients of $e_k\star e_r$ are regular at $t=0$, and*

$$(e_k\star e_r)\big|_{t=0}=\sum_{b=0}^{\mu-1}(1-s)s^b\,e_b\,e_{r+k-b}+s^\mu\,e_\mu e_M.$$

For $s\in[0,1]$ the weights $(1-s)s^b$ ($b<\mu$) and s^μ are the law of $\min(G,\mu)$ for a geometric variable G with $P(G=b)=(1-s)s^b$, and the monomials e_be_{r+k-b} ($b<\mu$), $e_\mu e_M$ are pairwise distinct. So at $t=0$, $e_k\star e_r$ is a truncated-geometric average of ordinary products.

Corollary 3.7 (Two columns; bounded support)**.** *For $\ell=2$, Theorem 3.2 reads*

$$\sum_{a,b\ge 0}z^aw^b\;e_k\star(e_ae_b)=\sum_{b_0,i,j}s^{2b_0}t^{-b_0(i+j)}\,V_{ij}^{(k-b_0)}\,e_{b_0}E(t^iz)E(t^jw),$$

$$V_{ij}^{(n)}=K_{ij}(z,w)\sum_{n_1+n_2=n}s^{n-i-j}\,t^{-(n_1-i)j-(n_2-j)i-n_1i-n_2j}\,c(n_1,i)\,c(n_2,j)\,z^{-n_1}w^{-n_2}.$$

In general, $e_k\star(e_{a_1}\cdots e_{a_\ell})$ is a linear combination of products of at most $\ell+1$ elementary symmetric functions.

Proof. The first part is the case $\ell=2$. For the second, expand both sides of Theorem 3.2 in the field of iterated Laurent series $\mathbb{Q}(s,t)((z_\ell))\cdots((z_1))$. The left side is a polynomial, so its expansion is itself. On the right, each term is $R(z)\,e_b\prod_c E(t^{i_c}z_c)$ with R rational and free of x, and its coefficient of z^a is a finite $\mathbb{Q}(s,t)$-combination of products $e_b\prod_{c=1}^{\ell}e_{n_c}$. □

3.3. **The recursion.** By Theorem 2.5, $e_k\star(e_{a_1}\cdots e_{a_\ell})=\sum_{|A|=k}c_Ax_A\prod_c e_{a_c}(x_{A^c},sx_A)$, and $\sum_a z^a\prod_c e_{a_c}(x_{A^c},sx_A)=\prod_c\prod_{j\notin A}(1+x_jz_c)\prod_{i\in A}(1+sx_iz_c)$. Hence

$$\mathcal{E}_k(x)=\prod_c E(z_c)\cdot S_k^{\varphi}(x),\qquad \varphi(u):=u\prod_{c=1}^{\ell}\frac{1+suz_c}{1+uz_c}. \tag{3.1}$$

Since $\varphi(x_i)\prod_c E(z_c)=x_i\prod_c(1+sx_iz_c)\cdot\prod_c E(\hat{x}_i;z_c)$, Lemma 2.9 gives, for $m\ge 1$ and $k\ge 1$,

$$[k]\,\mathcal{E}_k(x)=\sum_{i=1}^{m}x_i\prod_{c=1}^{\ell}(1+sx_iz_c)\prod_{j\ne i}a_{ij}\;\mathcal{E}_{k-1}(\hat{x}_i), \tag{3.2}$$

where $\mathcal{E}_{k-1}(\hat{x}_i)$ denotes $\mathcal{E}_{k-1}$ computed in the $m-1$ variables $\hat{x}_i$. For $m=0$ we have $\mathcal{E}_0=1$ and $\mathcal{E}_k=0$ for $k\ge 1$.

3.4. **The step map.** Fix I and put $\gamma_c:=t^{i_c}z_c$, $C:=s^\ell t^{-|I|}=s^\ell\prod_c z_c/\prod_c\gamma_c$, and

$$\mathcal{F}(v):=\frac{\prod_{c=1}^{\ell}(v-sz_c)}{v\prod_{c=1}^{\ell}(v-\gamma_c)}=\frac{C}{v}+\sum_{c=1}^{\ell}\frac{\eta_c}{v-\gamma_c},\qquad \eta_c:=\frac{\prod_{c'}(\gamma_c-sz_{c'})}{\gamma_c\prod_{c'\ne c}(\gamma_c-\gamma_{c'})}.$$

Lemma 3.8 (Step map)**.** *For $b\ge 0$, in $m\ge 1$ variables,*

$$\begin{aligned}(1-t)\sum_i x_i\prod_c(1+sx_iz_c)\prod_{j\ne i}a_{ij}\;e_b(\hat{x}_i)\prod_c E(\hat{x}_i;\gamma_c)&=C(1-t^{b+1})\,e_{b+1}\prod_c E(\gamma_c)\\&\quad-\sum_{n=0}^{b}e_n\sum_{c=1}^{\ell}\eta_c\,\gamma_c^{n-b-1}\big(E(t\gamma_c)-t^nE(\gamma_c)\big)\prod_{c'\ne c}E(\gamma_{c'}).\end{aligned}$$

Proof. Write $u = x_i$. Then $e_b(\hat{x}_i) = [y^b]E(y)/(1+uy)$ and $E(\hat{x}_i;\gamma) = E(\gamma)/(1+\gamma u)$, so the left side is $\prod_c E(\gamma_c)\cdot[y^b]\,E(y)\,(1-t)\sum_i H(x_i;y)\prod_{j\neq i}a_{ij}$ with

$$H(v;y) := \frac{v\prod_c(1+svz_c)}{(1+yv)\prod_c(1+\gamma_c v)}$$

(the sum over i is finite and each term is regular at $y=0$, so $[y^b]$ commutes with it). Apply Lemma 2.10 over $\mathbb{K} = \mathbb{Q}(s,t,z,y)$. The poles of H/v are simple, at $v=-1/a$ for $a\in\{y,\gamma_1,\dots,\gamma_\ell\}$, and $H(v)/v \sim C/(yv)$ at ∞, which contributes C/y. At $v=-1/a$ the residue of f is $\rho_a\,E(ta)/E(a)$, where

$$\rho_a = \frac{\prod_c(a-sz_c)}{a\prod_{a'\neq a}(a-a')},$$

as one sees from $\prod_c(1-sz_c/a) = a^{-\ell}\prod_c(a-sz_c)$, $\prod_{a'\neq a}(1-a'/a) = a^{-\ell}\prod_{a'\neq a}(a-a')$ and $Q(-a) = E(ta)/E(a)$. Thus $\rho_y = \mathcal{F}(y)$ and $\rho_{\gamma_c} = -\eta_c/(y-\gamma_c)$, and

$$(1-t)\sum_i H(x_i;y)\prod_{j\neq i}a_{ij} = \frac{C}{y} - \mathcal{F}(y)\frac{E(ty)}{E(y)} + \sum_c \frac{\eta_c}{y-\gamma_c}\,\frac{E(t\gamma_c)}{E(\gamma_c)}.$$

Multiply by $E(y)\prod_c E(\gamma_c)$ and write $E(y)=\sum_n e_n y^n$, $E(ty)=\sum_n t^n e_n y^n$. The coefficient of e_n is $[y^{b-n}]$ of the Laurent expansion at $y=0$ of

$$\Big(\frac{C}{y} - t^n\mathcal{F}(y)\Big)\prod_c E(\gamma_c) + \sum_c \frac{\eta_c}{y-\gamma_c}E(t\gamma_c)\prod_{c'\neq c}E(\gamma_{c'})$$
$$= \frac{C(1-t^n)}{y}\prod_c E(\gamma_c) + \sum_c \frac{\eta_c}{y-\gamma_c}\big(E(t\gamma_c) - t^n E(\gamma_c)\big)\prod_{c'\neq c}E(\gamma_{c'}),$$

by the partial fraction expansion of $\mathcal{F}$. Expanding $1/(y-\gamma) = -\sum_{p\geq 0} y^p\gamma^{-p-1}$ gives the claim: $n = b+1$ picks the pole at 0, $n\leq b$ picks the second sum, and $n > b+1$ contributes nothing. □

3.5. **Proof of Theorem 3.2.** Write $\mathcal{T}_k$ for the right-hand side of Theorem 3.2, so $\mathcal{T}_k = \sum_{b,I} W_{b,I}^{(k)}\,e_b\prod_c E(t^{i_c}z_c)$ with $W_{b,I}^{(k)} = C_I^b V_I^{(k-b)}$, $C_I := s^\ell t^{-|I|}$; $\epsilon_1,\dots,\epsilon_\ell$ are the unit vectors of $\mathbb{Z}^\ell$.

Reduction to an index identity. We show the step identity

$$\sum_i x_i\prod_c(1+sx_iz_c)\prod_{j\neq i}a_{ij}\,\mathcal{T}_{k-1}(\hat{x}_i) = [k]\,\mathcal{T}_k(x) \qquad (m\geq 0,\ k\geq 1) \tag{3.3}$$

(for $m=0$ the left side is empty). Given (3.3), the theorem follows by induction on k for all m at once: $\mathcal{E}_0 = \prod_c E(z_c) = \mathcal{T}_0$ because $V_0^{(0)} = 1$; for $m=0$, (3.3) says $\mathcal{T}_k = 0 = \mathcal{E}_k$; for $m\geq 1$, (3.2), the induction hypothesis in the variables $\hat{x}_i$, and (3.3) give $[k]\mathcal{E}_k = [k]\mathcal{T}_k$.

Apply Lemma 3.8 to each term of $\mathcal{T}_{k-1}(\hat{x}_i)$. The term $e_{b'}\prod_c E(t^{i_c}z_c)$ on the right receives three kinds of contributions: from $(b'-1, I)$ via the pole at 0; from (b, I), $b\geq b'$, via $-t^{b'}E(\gamma_c)$; and from $(b, I-\epsilon_c)$, $b\geq b'$, via $E(t\gamma_c')$ with $\gamma_c' = t^{i_c-1}z_c$. Writing η_c' for η_c computed at $I-\epsilon_c$, $n := k-b'$, $p := b-b'$, and using $C_{I-\epsilon_c} = tC_I$, the coefficient of $e_{b'}\prod_c E(t^{i_c}z_c)$ in $(1-t)$ times the left side of (3.3) is $C_I^{b'}$ times

$$(1-t^{b'})V_I^{(n)} + t^{b'}\sum_{p\geq 0} C_I^p\Big[V_I^{(n-1-p)}\sum_c \eta_c\gamma_c^{-p-1} - t^p\sum_c V_{I-\epsilon_c}^{(n-1-p)}\eta_c'\,(t^{i_c-1}z_c)^{-p-1}\Big].$$

So (3.3) follows from the index identity

$$\sum_{p\geq 0} C_I^p\Big[V_I^{(n-1-p)}\sum_c \eta_c\gamma_c^{-p-1} - t^p\sum_c V_{I-\epsilon_c}^{(n-1-p)}\eta_c'\,(t^{i_c-1}z_c)^{-p-1}\Big] = (1-t^n)V_I^{(n)} \qquad (n\geq 0), \tag{3.4}$$

because then the coefficient is $C_I^{b'}(1-t^{b'+n})V_I^{(n)} = (1-t^k)W_{b',I}^{(k)}$. (No uniqueness of such expansions is needed: the two sides agree term by term.)

Generating function. Let $\mathcal{V}_I(x) := \sum_{n\ge 0} V_I^{(n)} x^n$. The p-sums in (3.4) are convolutions, and $\sum_{p\ge 0} C^p \gamma^{-p-1} x^{p+1} = x/(\gamma - Cx)$. So (3.4) for all n is equivalent to

$$\mathcal{V}_I(x)\,U(x) - \mathcal{V}_I(tx) = -\sum_c \eta_c' \,\frac{x}{t^{i_c-1}z_c - C_I t x}\,\mathcal{V}_{I-\epsilon_c}(x), \qquad U(x) := 1 - \sum_c \frac{\eta_c x}{\gamma_c - C_I x}. \tag{3.5}$$

Put $X_c := C_I x/(s z_c)$ and $A_c := t^{i_c}$. Summing over (n_c) in the definition of $V_I^{(n)}$, the exponent of t attached to column c is $-n_c|I| + i_c(|I|-i_c)$, so

$$\mathcal{V}_I(x) = K_I\, s^{-(\ell-1)|I|}\, t^{|I|^2 - \sum_c i_c^2} \prod_c C_{i_c}(X_c), \tag{3.6}$$

and passing from I to $I-\epsilon_c$ replaces every $X_{c'}$ by $tX_{c'}$.

Step A: U is a product. By the partial fraction expansion of $\mathcal{F}$, $U(x) = x\,\mathcal{F}(C_I x)$. Since $C_I x - s z_c = -s z_c(1-X_c)$, $C_I x - \gamma_c = -\gamma_c(1 - sX_c/A_c)$ and $C_I \prod_c \gamma_c = s^\ell \prod_c z_c$,

$$U(x) = \prod_{c=1}^{\ell} \frac{1-X_c}{1 - sX_c/A_c}.$$

Step B: normalisation. Divide (3.5) by $\mathcal{N} := K_I\, s^{-(\ell-1)|I|} t^{|I|^2 - \sum i_c^2} \prod_c D_{i_c} X_c^{i_c} G(t^2 X_c)$, which is legitimate since $D_j \neq 0$. By Lemma 3.1 and $G(tX)/G(t^2X) = (1-stX)/(1-tX)$, with $u_c := stX_c$:

- column c of $\mathcal{V}_I U$ contributes $\frac{(1-stX_c)(1-sX_c/A_c)}{(1-tX_c)(1-X_c)} \cdot \frac{1-X_c}{1-sX_c/A_c} = \frac{1-u_c}{1-tX_c}$, and column c of $\mathcal{V}_I(tx)$ contributes $\frac{A_c - u_c}{1-tX_c}$;
- in the c-th term on the right (present only if $i_c \ge 1$), the columns $c' \neq c$ contribute $\frac{A_{c'} - u_{c'}}{1-tX_{c'}}$; column c contributes $\frac{D_{i_c-1}}{D_{i_c}} \frac{A_c}{t} X_c^{-1} \frac{1 - st^2 X_c/A_c}{1-tX_c}$ with $D_{i-1}/D_i = (1-A)/(ts-A)$; the prefactors contribute $s^{\ell-1} t^{-2(|I|-i_c)}$; and $\frac{x}{t^{i_c-1}z_c - C_I t x} = \frac{stX_c}{C_I(A_c - st^2 X_c)}$.

It remains to compute $\Phi_c := \eta_c' K_{I-\epsilon_c}/K_I$. The diagonal factor of η_c' is $(\gamma_c' - s z_c)/\gamma_c' = (A_c - st)/A_c$. For $c' \neq c$ put $z = z_c$, $w = z_{c'}$, $i = i_c$, $j = i_{c'}$, using $K_{ij}(z,w) = K_{ji}(w,z)$ to place c in the first slot. Then

$$\frac{t^{i-1}z - sw}{t^{i-1}z - t^j w} \cdot \frac{K_{i-1,j}(z,w)}{K_{ij}(z,w)} = \prod_{r=0}^{j-1} \frac{t^i z - t^r w}{t^{i-1}z - t^r w} = t^j\,\frac{t^i z - w}{t^i z - t^j w},$$

the last product telescoping after writing $t^i z - t^r w = t(t^{i-1}z - t^{r-1}w)$. With $\rho_{cc'} := z_c/z_{c'} = X_{c'}/X_c$ we get

$$\Phi_c = \frac{A_c - st}{A_c} \prod_{c'\neq c} A_{c'} \frac{A_c \rho_{cc'} - 1}{A_c \rho_{cc'} - A_{c'}}.$$

Multiplying everything, $\frac{stX_c}{C_I(A_c - st^2X_c)} \cdot \frac{A_c}{t} X_c^{-1}(1 - st^2 X_c/A_c) = s/C_I$, then $\frac{s}{C_I} s^{\ell-1} t^{-2(|I|-i_c)} = t^{2i_c - |I|} = A_c / \prod_{c'\neq c} A_{c'}$, and $(A_c - st)/(ts - A_c) = -1$. Hence, after division by $\mathcal{N}$ and multiplication by $\prod_c (1 - tX_c)$, (3.5) becomes

$$\prod_c (1-u_c) - \prod_c (A_c - u_c) - \sum_c (1 - A_c) \prod_{c'\neq c} \frac{A_c X_{c'} - X_c}{A_c X_{c'} - A_{c'} X_c}\,(A_{c'} - u_{c'}) = 0. \tag{3.7}$$

(For $i_c = 0$ the c-th term is absent, consistently with the factor $1 - A_c = 0$.)

Step C: the closing identity. Put $\lambda_c := A_c/u_c$ and $\mu_c := 1/u_c$. Since $X_c \propto u_c$, $\frac{A_c X_{c'} - X_c}{A_c X_{c'} - A_{c'} X_c} = \frac{\lambda_c - \mu_{c'}}{\lambda_c - \lambda_{c'}}$, and $(A_{c'} - u_{c'})/u_{c'} = \lambda_{c'} - 1$, $(1 - A_c)/u_c = \mu_c - \lambda_c$. Dividing (3.7) by $\prod_c u_c$ turns it

into the case of Lemma 3.9 below. As rational functions of x, the $\lambda_c = t^{i_c} z_c/(tC_I x)$ are pairwise distinct and different from 1. Hence (3.5) holds, so (3.4) holds for every n, which proves (3.3) and the theorem. □

Lemma 3.9. *For indeterminates $\lambda_1, \dots, \lambda_\ell, \mu_1, \dots, \mu_\ell$,*

$$\prod_c (\mu_c - 1) - \prod_c (\lambda_c - 1) = \sum_{c=1}^{\ell} (\mu_c - \lambda_c) \prod_{c' \neq c} (\lambda_{c'} - 1) \frac{\lambda_c - \mu_{c'}}{\lambda_c - \lambda_{c'}}.$$

Proof. The rational function

$$g(w) = \prod_c (w - \mu_c) \Big/ \Big((w-1) \prod_c (w - \lambda_c) \Big)$$

has residue $\frac{\lambda_c - \mu_c}{\lambda_c - 1} \prod_{c' \neq c} \frac{\lambda_c - \mu_{c'}}{\lambda_c - \lambda_{c'}}$ at λ_c, residue $\prod_c (1-\mu_c)/\prod_c (1-\lambda_c)$ at 1, and residue -1 at ∞. The residues sum to zero; multiply by $\prod_c (\lambda_c - 1)$. □

Remark 3.10. For $\ell = 1$, Lemma 3.9 is the tautology $(\mu - 1) - (\lambda - 1) = \mu - \lambda$, and the proof above reduces to a one-index recursion for $c(n, j)$. For $\ell = 2$ it is the polynomial identity that closes the two-column case. The cross kernel K_I is forced by the pairwise splitting of η_c' in Step B.

3.6. **Proof of Corollary 3.6.** At $t = 0$, $(t;t)_n = 1$, so every α_j and $c(n, j)$ is regular at $t = 0$; $(s;t)_n|_{t=0} = 1 - s$ for $n \geq 1$; hence $F_n(0) = c(n, 0)|_{t=0} = 1 - s$ for $n \geq 1$, and $F_0 = 1$.

Case $r \geq k$. Every $b \in [0, k]$ has $r - b \geq 0$, so each term $s^b F_{k-b}(t^{r-b}) = s^b \sum_j c(k-b, j) t^{j(r-b)}$ of Corollary 3.3 is regular at $t = 0$. If $b < r$, the terms with $j \geq 1$ vanish at $t = 0$, leaving $(1 - s)s^b$ for $b < k$ and s^k for $b = k$. If $b = r$, then $b = k = r$ and the term is $s^k F_0(1) = s^k$. Summing gives the claim with $\mu = k$.

Case $r < k$. By commutativity of $\star$ [Hik25], $e_k \star e_r = e_r \star e_k$. For $r = 0$ the claim is $e_k \star 1 = e_k$; for $r \geq 1$ apply the first case to (r, k). □

In the (k, r)-expansion of Corollary 3.3 with $r < k$, the terms with $b > r$ carry negative powers $t^{j(r-b)}$; the argument shows that these cancel once each monomial $e_b e_{r+k-b}$ is collected with its partner $b' = r + k - b$.

Remark 3.11 (Other forms and earlier cases). A second, independent proof of the Pieri rule, in the form $e_k \star e_r = \sum_y s^y \mathcal{C}_{k-y, r-y} e_{k+r-y} e_y$ with $\mathcal{C}_{a,b} = (-1)^b \frac{[a+b]}{[a]} \pi_a(b)$, is given in Section 8 (Corollary 8.6); it goes through the plethystic linear coefficient of Theorem 8.5 and the Column Lemma. The two forms agree after collecting terms. The kernel form of $e_k(Y)$ on symmetric functions is classical in spirit (Macdonald's D-operators); what is new here is the level-one transfer π^k, the explicit coefficients, and the ℓ-column rule.

4. Dominance support

For a partition λ write

$$e_\lambda^\star := e_{\lambda_1} \star \cdots \star e_{\lambda_\ell} = E_{\lambda_1} \cdots E_{\lambda_\ell}(1) = \sum_\mu c_{\lambda\mu}(s, t)\, e_\mu,$$

with E_k as in Theorem 2.5. The proofs below apply the operators E_{λ_i} in an arbitrary order, so they use neither commutativity nor associativity of $\star$.

Theorem 4.1 (Dominance support; the $s = 0$ valuation)**.** *$e_\lambda^\star = s^{n(\lambda)} e_\lambda + \sum_{\mu \rhd \lambda} c_{\lambda\mu} e_\mu$ with $c_{\lambda\mu} \in \mathbb{Q}[s, t]$ and $c_{\lambda\mu}|_{s=1} = 0$. Moreover $c_{\lambda\mu} \neq 0$ for every $\mu \trianglerighteq \lambda$, and the s-adic valuation of $c_{\lambda\mu}$ is exactly $n(\mu)$. At $t = 0$ the s-leading coefficients $[s^{n(\mu)}]c_{\lambda\mu}$ are all 1: in the basis $b_\mu = s^{n(\mu)} e_\mu$, $e_\lambda^\star$ at $s = t = 0$ is the zeta function of dominance order.*

Corollary 4.2 (Operator form)**.** *For $k \geq 1$ and every partition μ, $e_k \star e_\mu \in s^{\sum_i \min(\mu_i,k)} e_{\mu\cup k} + \operatorname{span}\{e_\nu : \nu \triangleright \mu \cup k\}$.*

We work in Λ_m with $m \geq n = |\lambda|$; by Lemma 2.8 this loses nothing.

4.1. Degrees and dominance.

Lemma 4.3. *Let $m \geq n$ and $\rho \vdash n$. Then $\operatorname{span}\{e_\nu : \nu \trianglerighteq \rho'\} = \operatorname{span}\{m_\kappa : \kappa \trianglelefteq \rho\}$ in the degree-n part of Λ_m.*

Proof. By [Mac95, I (6.6), (6.7)], $e_\nu = \sum_\kappa M_{\nu\kappa} m_\kappa$, where $M_{\nu\kappa}$ is the number of 0-1 matrices with row sums ν and column sums κ; $M_{\nu\kappa} \neq 0$ forces $\kappa \trianglelefteq \nu'$, and $M_{\nu\nu'} = 1$. Since $\nu \trianglerighteq \rho' \iff \nu' \trianglelefteq \rho$ [Mac95, I (1.11)], the left side is contained in the right. Both have dimension $\#\{\kappa \vdash n : \kappa \trianglelefteq \rho\}$, because the e_ν, $\nu \vdash n$, are independent for $m \geq n$. □

For $S \subseteq [m]$ and nonzero $R \in \mathbb{Q}(s,t)(x)$, let $\deg_S R$ be the degree in u of $R|_{x_i \mapsto u x_i\,(i\in S)}$ (numerator degree minus denominator degree). It is additive on products and satisfies $\deg_S(R_1 + R_2) \leq \max(\deg_S R_1, \deg_S R_2)$. For a polynomial P, $\deg_S P$ is the largest value of $\sum_{i\in S} \alpha_i$ over the monomials x^α of P; for symmetric P it depends only on $j = |S|$, and we write $D_j(P)$. Since the m_κ have disjoint supports and $D_j(m_\kappa) = \kappa_1 + \cdots + \kappa_j$:

Lemma 4.4. *A homogeneous $P \in \Lambda_m$ of degree $|\rho|$ lies in $\operatorname{span}\{m_\kappa : \kappa \trianglelefteq \rho\}$ if and only if $D_j(P) \leq \rho_1 + \cdots + \rho_j$ for all j.*

Lemma 4.5 (Degree step)**.** *For symmetric F and every j, $D_j(E_k F) \leq D_j(F) + \min(k, j)$.*

Proof. Fix $|S| = j$ and a term $R_A = c_A\, x_A\, F|_{x_i \mapsto s x_i\,(i\in A)}$ of Theorem 2.5. Every a_{ij} has $\deg_S a_{ij} = 0$: if i, j are both in S or both outside, a_{ij} is homogeneous of degree 0 in u or constant; otherwise numerator and denominator have degree 1. Next, $\deg_S x_A = |A \cap S| \leq \min(k, j)$. Rescaling some variables by the constant s does not change u-degrees, so the last factor has $\deg_S = D_j(F)$. Since $E_k F$ is a polynomial, its $\deg_S$ is its monomial degree. □

4.2. Support, diagonal coefficient, and the value at $s = 1$.

Proof of the support statement. Starting from $D_j(1) = 0$ and applying Lemma 4.5 once for each part, $D_j(e^\star_\lambda) \leq \sum_i \min(\lambda_i, j) = \lambda'_1 + \cdots + \lambda'_j$. By Lemmas 4.3 and 4.4, $e^\star_\lambda \in \operatorname{span}\{m_\kappa : \kappa \trianglelefteq \lambda'\} = \operatorname{span}\{e_\nu : \nu \trianglerighteq \lambda\}$. The support part of Corollary 4.2 is the same argument with one step, starting from $D_j(e_\mu) = \sum_i \min(\mu_i, j)$. □

Fix integers $w_1 > w_2 > \cdots > w_m > 0$ and for $\rho = (\rho_1 \geq \cdots \geq \rho_m \geq 0)$ put $W(\rho) = \sum_i w_i \rho_i$.

Lemma 4.6. *If $\operatorname{sort}(\alpha) = \kappa \trianglelefteq \rho$, then $\sum_i w_i \alpha_i \leq W(\rho)$, with equality only if $\alpha = \rho$. Hence if $P \in \operatorname{span}\{m_\kappa : \kappa \trianglelefteq \rho\}$ has m_ρ-coefficient c, the coefficient of $u^{W(\rho)}$ in $P(u^{w_1}x_1, \ldots, u^{w_m}x_m)$ is $c\,x^\rho$, and no higher power of u occurs.*

Proof. $\sum_i w_i\alpha_i \leq \sum_i w_i\kappa_i$ by the rearrangement inequality, with equality only if $\alpha = \kappa$ (the w_i are distinct). By Abel summation, $\sum_i w_i\kappa_i = \sum_j (w_j - w_{j+1})(\kappa_1 + \cdots + \kappa_j)$ with $w_{m+1} := 0$; all weights are positive, so this is at most the same expression for ρ, with equality only if $\kappa = \rho$. □

Lemma 4.7 (Lead step)**.** *Let $F \in \operatorname{span}\{m_\kappa : \kappa \trianglelefteq \rho\}$ with m_ρ-coefficient c_F. Then $E_k F \in \operatorname{span}\{m_\kappa : \kappa \trianglelefteq \rho + 1^k\}$, with $m_{\rho+1^k}$-coefficient $s^{\rho_1+\cdots+\rho_k} c_F$.*

Proof. Membership is Lemmas 4.4 and 4.5. Substitute $x_i \mapsto u^{w_i} x_i$ and expand each term R_A in u^{-1}. For $i < j$, $a_{ij} = 1 + O(u^{-1})$, and for $i > j$, $a_{ij} = t + O(u^{-1})$, so $c_A = t^{\operatorname{inv}(A)}(1 + O(u^{-1}))$ with $\operatorname{inv}(A) = \#\{i \in A, j \notin A : i > j\}$. Next, x_A becomes $u^{\sum_{a\in A} w_a} x_A$. Finally, $F|_{x_a \mapsto s x_a\,(a\in A)}$ has the monomials of F with x^α rescaled by $s^{\sum_{a\in A}\alpha_a}$, so by Lemma 4.6 it is $u^{W(\rho)}\big(c_F s^{\sum_{a\in A}\rho_a} x^\rho +$

$O(u^{-1})$). The exponent $W(\rho)+\sum_{a\in A} w_a$ is uniquely maximal for $A=[k]$, where inv $=0$. So the coefficient of $u^{W(\rho+1^k)}$ in E_kF is $s^{\rho_1+\cdots+\rho_k}c_F\,x^{\rho+1^k}$; conclude by Lemma 4.6 for $\rho+1^k$. □

Proof of the diagonal coefficient. Apply the operators in any order. If the parts applied so far form μ, the current function lies in $\operatorname{span}\{m_\kappa : \kappa \trianglelefteq \mu'\}$, and applying E_k multiplies the $m_{\mu'}$-coefficient by $s^{\mu'_1+\cdots+\mu'_k} = s^{\sum_i \min(\mu_i,k)}$ (Lemma 4.7; note $\mu'+1^k=(\mu\cup k)'$). Each unordered pair of parts contributes its minimum once, so the total exponent is $\sum_{i<p}\min(\lambda_i,\lambda_p)=\sum_p(p-1)\lambda_p = n(\lambda)$. By the triangularity in the proof of Lemma 4.3, the only e_ν with $\nu\trianglerighteq\lambda$ containing $m_{\lambda'}$ is e_λ, with coefficient 1. Hence $c_{\lambda\lambda}=s^{n(\lambda)}$. The same computation gives the leading term in Corollary 4.2. □

Proof of integrality and of $c_{\lambda\mu}|_{s=1}=0$. Let $V=\prod_{i<j}(x_i-x_j)$. Multiplying the subset formula by V gives

$$V\,E_kF=\sum_{|A|=k}\varepsilon_A \prod_{\substack{i<j\\ \text{both in or both out of } A}}(x_i-x_j)\prod_{i\in A,\,j\notin A}(x_i-tx_j)\;x_A\,F|_{x_a\mapsto sx_a\,(a\in A)},\qquad \varepsilon_A=\pm1.$$

If $F\in\mathbb{Q}[s,t][x]$, the right side lies in $\mathbb{Q}[s,t][x]$. As V has content 1 and E_kF is a polynomial, Gauss's lemma gives $E_kF\in\mathbb{Q}[s,t][x]$. By induction $e^\star_\lambda\in\mathbb{Q}[s,t][x]^{S_m}$, and its e-coefficients lie in $\mathbb{Q}[s,t]$. Now set $s=1$ (a ring map on $\mathbb{Q}[s,t][x]$): every $F|_{x_a\mapsto sx_a}$ becomes $F|_{s=1}$, so $V\,(E_kF)|_{s=1}=F|_{s=1}\cdot V\,E_k(1)=V\,e_k\,F|_{s=1}$ by Lemma 2.7. Hence $(E_kF)|_{s=1}=e_kF|_{s=1}$, $e^\star_\lambda|_{s=1}=e_\lambda$, and $c_{\lambda\mu}|_{s=1}=\delta_{\lambda\mu}$. □

4.3. **The exact s-adic valuation.** It remains to show that $s^{n(\mu)}$ divides $c_{\lambda\mu}$ and that $(c_{\lambda\mu}/s^{n(\mu)})|_{s=t=0}=1$ for all $\mu\trianglerighteq\lambda$; in particular $c_{\lambda\mu}\neq0$. Put $\omega(\alpha):=\sum_j\binom{\alpha_j}{2}$ for $\alpha\in\mathbb{Z}^m_{\geq0}$ and $B_\mu:=\{\beta:\operatorname{sort}(\beta)\trianglelefteq\mu'\}$, the monomial support of $e^\star_\mu$. Write $e^\star_\lambda=\sum_\beta a^\lambda_\beta x^\beta$; a^λ_β is symmetric in β. The source of s^n is the identity

$$\omega(\beta+1_A)=\omega(\beta)+1_A\cdot\beta. \tag{4.1}$$

Let $\mathbb{K}=\mathbb{Q}(t)(s^{1/2})$ (or $\mathbb{Q}(s^{1/2})$ when $t=0$) with the s-adic valuation v, extended to $\mathbb{K}(x)$ by the Gauss valuation $v(\sum a_\beta x^\beta)=\min_\beta v(a_\beta)$. For $c\in(\frac12\mathbb{Z})^m$ let σ_c be the automorphism $x_j\mapsto s^{-c_j}x_j$, so that $v(\sigma_cP)=\min_\beta(v(a_\beta)-c\cdot\beta)$.

Lemma 4.8. *For generic t, and also at $t=0$, $v(a^\lambda_\alpha)\geq\omega(\alpha)$ for all α.*

Proof. We show $v(\sigma_ce^\star_\lambda)\geq\min_{\beta\in B_\lambda}(\omega(\beta)-c\cdot\beta)$ for all c, by induction along $e^\star_\mu\mapsto E_ke^\star_\mu=e^\star_{\mu\cup k}$, the case $e^\star_\varnothing=1$ being clear. In a term R_A: $v(\sigma_ca_{ij})=0$ for generic t, and at $t=0$, $v(\sigma_ca_{ij})=\max(c_i,c_j)-c_i\geq0$; $\sigma_cx_A=s^{-c\cdot1_A}x_A$; and $\sigma_c(F|_{x_a\mapsto sx_a\,(a\in A)})=\sigma_{c-1_A}F$. Hence, by induction and (4.1),

$$v(\sigma_cR_A)\geq-c\cdot1_A+\min_{\beta\in B_\mu}\big(\omega(\beta)-(c-1_A)\cdot\beta\big)=\min_{\beta\in B_\mu}\big(\omega(\beta+1_A)-c\cdot(\beta+1_A)\big),$$

and $\beta+1_A\in B_{\mu\cup k}$ because the top-j sums grow by at most $\min(j,k)$ and $\mu'+1^k=(\mu\cup k)'$. Now take $c=\alpha-\frac12\mathbf{1}$. Then $\omega(\beta)-c\cdot\beta=\sum_j((\beta_j-\alpha_j)^2-\alpha_j^2)/2$ is uniquely minimised over $\mathbb{Z}^m$ at $\beta=\alpha$, so $v(a^\lambda_\alpha)-c\cdot\alpha\geq v(\sigma_ce^\star_\lambda)\geq\omega(\alpha)-c\cdot\alpha$. □

Proof that $s^{n(\mu)}\mid c_{\lambda\mu}$. Write $c_{\lambda\nu}=s^{n(\nu)}d_\nu$ and suppose $\delta:=\min_\nu v(d_\nu)<0$. Choose μ dominance-minimal with $v(d_\mu)=\delta$. The coefficient of $x^{\mu'}$ is $a^\lambda_{\mu'}=\sum_{\nu\trianglelefteq\mu}c_{\lambda\nu}M_{\nu\mu'}$ with $M_{\mu\mu'}=1$. Since $\nu\triangleleft\mu$ implies $n(\nu)>n(\mu)$ (as $n(\nu)=\sum_{i\geq1}(n-\nu_1-\cdots-\nu_i)$), every term with $\nu\neq\mu$ has valuation $>n(\mu)+\delta$, so $v(a^\lambda_{\mu'})=n(\mu)+\delta<n(\mu)=\omega(\mu')$, contradicting Lemma 4.8. So d_ν is regular at $s=0$; as $c_{\lambda\nu}\in\mathbb{Q}[s,t]$, $s^{n(\nu)}$ divides $c_{\lambda\nu}$ in $\mathbb{Q}[s,t]$. The same argument at $t=0$ shows $d_\mu|_{s=t=0}=[s^{\omega(\mu')}]\,a^\lambda_{\mu'}|_{t=0}$. □

For the value at $s = t = 0$ we need the initial forms. For $R \in \mathbb{K}(x)$ with $v(R) \geq v_0$, let $\mathrm{in}_{v_0}(R)$ be the image of $s^{-v_0}R$ in the residue field of $\mathbb{K}(x)$, which is $\mathbb{Q}(t)(x)$ (or $\mathbb{Q}(x)$ when $t = 0$); it is additive at a common level and multiplicative. For $\kappa = \mathrm{sort}(\alpha)$ let peel_k subtract 1 from the k largest parts.

Lemma 4.9 (Peel Lemma)**.** *If $\kappa \trianglelefteq \rho$, $|\kappa| = |\rho|$ and $k \leq \ell(\rho)$, then $\mathrm{peel}_k\kappa \trianglelefteq \mathrm{peel}_k\rho$.*

Proof. Note $\ell(\kappa) \geq \ell(\rho) \geq k$. Counting parts $\geq c$ gives $(\mathrm{peel}_k\kappa)'_c = \kappa'_c - \min(k, \kappa'_c) + \min(k, \kappa'_{c+1})$, so with S_C the sum of the first C parts and $\kappa'_1 \geq k$, $S_C((\mathrm{peel}_k\kappa)') = S_C(\kappa') - k + \min(k, \kappa'_{C+1})$, and likewise for ρ. Let $D_C = S_C(\kappa') - S_C(\rho') \geq 0$ (by [Mac95, I (1.11)]) and $\bar{m}(y) = \min(k, y)$. If $\bar{m}(\kappa'_{C+1}) \geq \bar{m}(\rho'_{C+1})$ we are done. Otherwise $\kappa'_{C+1} < k$ and $\kappa'_{C+1} < \rho'_{C+1}$, and $D_C = D_{C+1} + \rho'_{C+1} - \kappa'_{C+1} \geq \bar{m}(\rho'_{C+1}) - \bar{m}(\kappa'_{C+1})$. So $(\mathrm{peel}_k\kappa)' \trianglerighteq (\mathrm{peel}_k\rho)'$. □

Lemma 4.10. *At $t = 0$, $\Lambda^0_\lambda(\alpha) := [s^{\omega(\alpha)}]\, a^\lambda_\alpha$ equals 1 if $\mathrm{sort}(\alpha) \trianglelefteq \lambda'$ and 0 otherwise.*

Proof. Induction along E_k; the base case is clear, and $\Lambda^0_{\mu\cup k}(\alpha) = 0$ outside $B_{\mu\cup k}$ by the support statement. Let $\mathrm{sort}(\alpha) \trianglelefteq (\mu \cup k)'$, $c = \alpha - \frac{1}{2}\mathbf{1}$ and $v_0 = \omega(\alpha) - c \cdot \alpha$. By the uniqueness in the proof of Lemma 4.8, $\mathrm{in}_{v_0}(\sigma_c e^\star_{\mu\cup k}) = \Lambda^0_{\mu\cup k}(\alpha)\, x^\alpha$. If $\alpha - 1_A$ has a negative entry or lies outside B_μ, then $v(\sigma_c R_A) > v_0$ by that uniqueness and the term R_A does not contribute. Otherwise, with $\beta_0 = \alpha - 1_A$, the levels split as in (4.1) and

$$\mathrm{in}_{v_0}(\sigma_c R_A) = \mathrm{in}_0(\sigma_c c_A) \cdot \Lambda^0_\mu(\beta_0)\, x^\alpha.$$

At $t = 0$, $\mathrm{in}_0(\sigma_c a_{ij})$ is 1 if $\alpha_i > \alpha_j$, 0 if $\alpha_i < \alpha_j$, and $x_i/(x_i - x_j)$ if $\alpha_i = \alpha_j$. So only sets A that are upper sets for α contribute: with v the k-th largest entry of α and $L = \{j : \alpha_j = v\}$, $A = \{j : \alpha_j > v\} \sqcup B$ with $B \subseteq L$ of size $r = k - \#\{j : \alpha_j > v\}$. As $\ell(\mathrm{sort}\,\alpha) \geq \ell((\mu \cup k)') \geq k$, $v \geq 1$. For all such A, $\mathrm{sort}(\alpha - 1_A) = \mathrm{peel}_k(\mathrm{sort}\,\alpha)$, and Λ^0_μ is symmetric, so

$$\Lambda^0_{\mu\cup k}(\alpha) = \Lambda^0_\mu(\mathrm{peel}_k\,\mathrm{sort}\,\alpha) \cdot \sum_{B\subseteq L,\ |B|=r} \prod_{i\in B,\ j\in L\setminus B} \frac{x_i}{x_i - x_j} = \Lambda^0_\mu(\mathrm{peel}_k\,\mathrm{sort}\,\alpha).$$

The last sum is the case $t = 0$ of $\sum_{|B|=r} \prod_{i\in B, j\in L\setminus B} \frac{x_i - t x_j}{x_i - x_j} = \begin{bmatrix} |L| \\ r \end{bmatrix}_t$, which follows from the Poincaré identity [Mac95, III (1.4)] by grouping $S_{|L|}$ according to the image of $\{1, \dots, r\}$. Finally, by Lemma 4.9 with $\rho = (\mu \cup k)' = \mu' + 1^k$, $\mathrm{peel}_k(\mathrm{sort}\,\alpha) \trianglelefteq \mathrm{peel}_k(\mu' + 1^k) = \mu'$, so $\Lambda^0_\mu(\mathrm{peel}_k\,\mathrm{sort}\,\alpha) = 1$. □

End of proof of Theorem 4.1. By the last two proofs, $(c_{\lambda\mu}/s^{n(\mu)})|_{s=t=0} = \Lambda^0_\lambda(\mu') = 1$ for $\mu \trianglerighteq \lambda$, since $\mu' \trianglelefteq \lambda'$. In particular $c_{\lambda\mu} \neq 0$ and its s-adic valuation is $n(\mu)$. □

Remark 4.11. The leading coefficients $d_{\lambda\mu}(t) = [s^{n(\mu)}]c_{\lambda\mu}$ depend on t; for instance $d_{(1^4),(2,1,1)} = 1 + 3t$. They are identified in Corollary 5.2. The degree count above is the same kind of triangularity argument as for Macdonald's operators [Mac95, VI (3.6)–(3.10)].

5. The $s \to 0$ edge

Since $s = q^{-1}$, the edge $s \to 0$ is Hikita's $q \to \infty$, where the unrescaled product degenerates [Hik25, Lemma 6.3]. After the rescaling of Theorem 4.1 it becomes Hall–Littlewood multiplication. Throughout this section t is an indeterminate, $b_\mu := s^{n(\mu)}e_\mu$, and we work in Λ_m with m at least the degree (Lemma 2.8). We keep the notation $\mathbb{K}$, v, σ_c, ω, in_{v_0} of Section 4, with $\mathbb{K} = \mathbb{Q}(t)(s^{1/2})$; so initial forms take values in $\mathbb{Q}(t)(x)$.

Theorem 5.1 (The $s \to 0$ edge)**.** *Let $\mathcal{O} = \mathbb{Q}(t)[s]_{(s)}$ and $\mathcal{L} = \bigoplus_\mu \mathcal{O}\, b_\mu$.*

(1) *$E_k b_\mu \in \bigoplus_\nu \mathbb{Q}[s,t]\, b_\nu$; in particular $E_k\mathcal{L} \subseteq \mathcal{L}$.*

(2) *Let L_k be the map induced by E_k on the quotient $\mathcal{L}/s\mathcal{L} = \bigoplus_\mu \mathbb{Q}(t)\,\bar{b}_\mu$, and let $\phi(\bar{b}_\mu) := t^{-n(\mu')}P_{\mu'}(x;t^{-1})$. Then $\phi \circ L_k = t^{-\binom{k}{2}}\, e_k \cdot \phi$.*

(3) *Explicitly, with $\rho = \mu'$ and $\kappa = \nu'$,*

$$L_k\bar{b}_\mu = \sum_{\kappa/\rho \text{ vertical } k\text{-strip}} t^{\mathrm{inv}(\kappa/\rho)} \prod_{v\geq 1} \begin{bmatrix} m_v(\kappa) \\ r_v \end{bmatrix}_t \bar{b}_\nu,$$

where r_v is the number of rows of κ of length v that end in a box of the strip, and $\mathrm{inv}(\kappa/\rho) = \sum_{v<w} r_v\,(m_w(\kappa) - r_w)$.

Corollary 5.2 (The s-leading coefficients)**.** *For $\lambda, \mu \vdash n$, $d_{\lambda\mu}(t) := [s^{n(\mu)}]\, c_{\lambda\mu}$ satisfies*

$$d_{\lambda\mu}(t) = t^{\,n(\mu')-n(\lambda')}\,[P_{\mu'}(x;t^{-1})]\,e_\lambda = t^{-n(\lambda')}\sum_\nu K_{\nu'\lambda}\,\widetilde{K}_{\nu\mu'}(t),$$

where $\widetilde{K}_{\nu\kappa}(t) = t^{n(\kappa)}K_{\nu\kappa}(t^{-1})$ is the cocharge Kostka–Foulkes polynomial. Hence $d_{\lambda\mu} \in \mathbb{N}[t]$, and $d_{\lambda\mu}(1)$ is the number of 0-1 matrices with row sums λ and column sums μ'.

5.1. **Admissible functions.** Call a homogeneous $F = \sum_\beta a_\beta x^\beta \in \mathbb{K}[x]^{S_m}$ *admissible* if $v(a_\beta) \geq \omega(\beta)$ for all β. As in the proof of Lemma 4.8, F is admissible if and only if $v(\sigma_c F) \geq \min_{\beta\in\mathbb{Z}^m_{\geq 0}}(\omega(\beta) - c\cdot\beta)$ for all $c \in (\frac{1}{2}\mathbb{Z})^m$; the backward direction uses $c = \alpha - \frac{1}{2}\mathbf{1}$, for which $\beta = \alpha$ is the unique minimiser. For admissible F let $\mathrm{in}(F)(\alpha) \in \mathbb{Q}(t)$ be the residue of $s^{-\omega(\alpha)}a_\alpha$; it depends only on $\mathrm{sort}(\alpha)$. With $c = \alpha - \frac{1}{2}\mathbf{1}$ and $v_0 = \omega(\alpha) - c\cdot\alpha$,

$$\mathrm{in}_{v_0}(\sigma_c F) = \mathrm{in}(F)(\alpha)\, x^\alpha. \tag{5.1}$$

Lemma 5.3. *A homogeneous $F = \sum_\mu c_\mu e_\mu$ of degree $\leq m$ is admissible if and only if $v(c_\mu) \geq n(\mu)$ for all μ. In that case $\mathrm{in}(F)(\mu')$ is the residue of $s^{-n(\mu)}c_\mu$.*

Proof. The coefficient of x^β in b_μ is $s^{n(\mu)}M_{\mu\kappa}$ with $\kappa = \mathrm{sort}(\beta)$. If $M_{\mu\kappa} \neq 0$ then $\kappa \trianglelefteq \mu'$, so $\kappa' \trianglerighteq \mu$ and $\omega(\beta) = n(\kappa') \leq n(\mu)$, with equality only for $\kappa = \mu'$, where $M_{\mu\mu'} = 1$. So b_μ is admissible with $\mathrm{in}(b_\mu)(\kappa) = [\kappa = \mu']$, which gives the backward direction and the formula. The forward direction is the argument in the proof that $s^{n(\mu)} \mid c_{\lambda\mu}$ in Section 4, with a^λ replaced by the coefficients of F. □

Lemma 5.4. $\sum_{B\subseteq[N],|B|=r}\prod_{i\in B, j\notin B} a_{ij} = \begin{bmatrix} N \\ r \end{bmatrix}_t$.

Proof. Group the Poincaré identity $\sum_{w\in S_N} w(\prod_{i<j} a_{ij}) = [N]_t!$ [Mac95, III (1.4)] by $B = w(\{1,\dots,r\})$; the stabiliser $S_r \times S_{N-r}$ contributes $[r]_t![N-r]_t!$. □

5.2. **The recursion.**

Proposition 5.5. *Let F be admissible of degree n, with $m \geq n + k$. Then E_kF is admissible, and for every $\kappa \vdash n + k$*

$$\mathrm{in}(E_kF)(\kappa) = \sum_{\rho:\ \kappa/\rho \text{ vertical } k\text{-strip}} t^{\mathrm{inv}(\kappa/\rho)} \prod_{v\geq 1} \begin{bmatrix} m_v(\kappa) \\ r_v \end{bmatrix}_t \mathrm{in}(F)(\rho). \tag{5.2}$$

Proof. Fix c and a term R_A of Theorem 2.5. Since t is a unit, $v(\sigma_c a_{ij}) = 0$, and the initial form of $\sigma_c a_{ij}$ is 1 if $c_i > c_j$, t if $c_i < c_j$, and a_{ij} if $c_i = c_j$. As in Lemma 4.8, using (4.1),

$$v(\sigma_c R_A) \geq \min_{\gamma\geq 1_A}(\omega(\gamma) - c\cdot\gamma) \geq \min_{\gamma\geq 0}(\omega(\gamma) - c\cdot\gamma),$$

so E_kF is admissible. Now fix α with $\mathrm{sort}(\alpha) = \kappa$, $c = \alpha - \frac{1}{2}\mathbf{1}$, $v_0 = \omega(\alpha) - c\cdot\alpha$; all terms have $v \geq v_0$, so initial forms add, and by (5.1) $\mathrm{in}(E_kF)(\alpha)x^\alpha = \sum_A \mathrm{in}_{v_0}(\sigma_c R_A)$. If $A \not\subseteq \mathrm{supp}(\alpha)$, then

$\gamma=\alpha$ is excluded from $\{\gamma\ge 1_A\}$ and $v(\sigma_c R_A)>v_0$. If $A\subseteq\operatorname{supp}(\alpha)$, put $\beta_0=\alpha-1_A$; then $c-1_A=\beta_0-\frac12\mathbf{1}$, the levels add by (4.1), and

$$\operatorname{in}_{v_0}(\sigma_c R_A)=\operatorname{in}_0(\sigma_c c_A)\cdot\operatorname{in}(F)(\beta_0)\,x^{\alpha}.$$

Let $L_v=\{j:\alpha_j=v\}$ $(v\ge1)$, so $|L_v|=m_v(\kappa)=:m_v$, and let $B_v=A\cap L_v$, $r_v=|B_v|$. Then

$$\operatorname{in}_0(\sigma_c c_A)=t^{\#\{i\in A,\,j\notin A:\ \alpha_i<\alpha_j\}}\prod_{v\ge1}\ \prod_{i\in B_v,\,j\in L_v\setminus B_v}a_{ij},\qquad \#\{\cdots\}=\sum_{v<w}r_v(m_w-r_w).$$

The partition $\operatorname{sort}(\beta_0)$ is obtained from κ by lowering r_v of its parts equal to v, for each v; this is the ρ with κ/ρ a vertical k-strip and data (r_v), and every such strip arises from exactly one (r_v). So both $\operatorname{in}(F)(\beta_0)$ and the power of t depend only on (r_v), and summing over the B_v with Lemma 5.4 gives $\prod_v\begin{bmatrix}m_v\\ r_v\end{bmatrix}_t$. □

At $t=0$ every t-binomial is 1 and t^{inv} vanishes unless every part larger than a lowered part is lowered; so (5.2) reduces to the recursion in the proof of Lemma 4.10.

5.3. **Proof of Theorem 5.1 and Corollary 5.2.** *(1).* b_μ is admissible, so $E_k b_\mu$ is admissible (Proposition 5.5) and its b_ν-coordinates have $v\ge0$ (Lemma 5.3). By the integrality part of Theorem 4.1's proof, $E_k e_\mu\in\bigoplus_\nu\mathbb{Q}[s,t]e_\nu$, so the b_ν-coordinate is $p/s^{n(\nu)-n(\mu)}$ with $p\in\mathbb{Q}[s,t]$; having $v\ge0$ over $\mathbb{Q}(t)$, it lies in $\mathbb{Q}[s,t]$.

(3). By Lemma 5.3, the $\bar b_\nu$-coordinate of $L_k\bar b_\mu$ is $\operatorname{in}(E_k b_\mu)(\nu')$; apply (5.2) with $\operatorname{in}(b_\mu)(\rho)=[\rho=\mu']$.

(2). Let κ/ρ be a vertical k-strip with data (r_v), obtained by lowering the parts indexed by A. Writing $n(\kappa)=\sum_{\text{pairs of parts}}\min$, a pair of parts both in A loses 1; a pair with one part of value v in A and one of value w not in A loses 1 if and only if $v\le w$; other pairs lose nothing. Hence

$$n(\kappa)-n(\rho)=\binom{k}{2}+\operatorname{inv}(\kappa/\rho)+\sum_v r_v(m_v-r_v). \tag{5.3}$$

Since $\begin{bmatrix}m\\ r\end{bmatrix}_t\big|_{t\mapsto t^{-1}}=t^{-r(m-r)}\begin{bmatrix}m\\ r\end{bmatrix}_t$, (5.3) gives $t^{\mathrm{inv}}\prod_v\begin{bmatrix}m_v\\ r_v\end{bmatrix}_t=t^{\,n(\kappa)-n(\rho)-\binom{k}{2}}\psi_{\kappa/\rho}(t^{-1})$ with $\psi_{\kappa/\rho}(u):=\prod_v\begin{bmatrix}m_v(\kappa)\\ r_v\end{bmatrix}_u$. The Hall–Littlewood Pieri rule [Mac95, III (3.2)] reads $e_kP_\rho(x;u)=\sum_\kappa\psi_{\kappa/\rho}(u)P_\kappa(x;u)$, summed over vertical k-strips (there $m_v(\kappa)=\kappa'_v-\kappa'_{v+1}$ and $r_v=\kappa'_v-\rho'_v$). For $X\in\mathcal{L}$ put $h(\kappa)=t^{-n(\kappa)}\operatorname{in}(X)(\kappa)$, so that $\phi(\bar X)=\sum_\kappa h(\kappa)P_\kappa(x;t^{-1})$ by Lemma 5.3. By (5.2), $t^{-n(\kappa)}\operatorname{in}(E_kX)(\kappa)=t^{-\binom{k}{2}}\sum_\rho\psi_{\kappa/\rho}(t^{-1})h(\rho)$, hence $\phi(L_k\bar X)=t^{-\binom{k}{2}}\sum_\rho h(\rho)\,e_kP_\rho(x;t^{-1})=t^{-\binom{k}{2}}e_k\,\phi(\bar X)$.

Corollary 5.2. Since $1=b_\varnothing$ and $\phi(\bar b_\varnothing)=1$, iterating (2) gives $\phi(e^\star_\lambda \bmod s)=t^{-\sum_i\binom{\lambda_i}{2}}e_\lambda=t^{-n(\lambda')}e_\lambda$. By Theorem 4.1, $e^\star_\lambda\equiv\sum_\mu d_{\lambda\mu}\bar b_\mu$, so comparing coefficients of $P_{\mu'}(x;t^{-1})$ gives the first formula. Expanding $e_\lambda=\omega h_\lambda=\sum_\nu K_{\nu'\lambda}s_\nu$ [Mac95, I §6] and $s_\nu=\sum_\kappa K_{\nu\kappa}(u)P_\kappa(x;u)$ [Mac95, III (2.6)] at $u=t^{-1}$ gives the second. The $K_{\nu\kappa}(t)$ have nonnegative coefficients [Mac95, III (6.5)], and $d_{\lambda\mu}\in\mathbb{Q}[t]$ by Theorem 4.1, so $d_{\lambda\mu}\in\mathbb{N}[t]$. At $t=1$, $P_\kappa(x;1)=m_\kappa$ [Mac95, III §2], so $d_{\lambda\mu}(1)=[m_{\mu'}]e_\lambda=M_{\lambda\mu'}$. □

Remark 5.6 (Prior work; no novelty claim for the matrix). The matrix $d_{\lambda\mu}(t)$ is the classical $e\to$Hall–Littlewood transition matrix: writing $e_\lambda=\sum_\kappa M(e,P)_{\lambda\kappa}P_\kappa(x;t)$, Corollary 5.2 says $d_{\lambda\mu}(t)=t^{\,n(\mu')-n(\lambda')}M(e,P)_{\lambda\mu'}(t^{-1})$, and its positivity and the value at $t=1$ are in [Kir98, §3.2]. What is new is only the identification of this matrix with the s-leading coefficients of $\star$, i.e. Theorem 5.1. Hikita's [Hik25, Lemma 6.3] treats the unrescaled limit $q\to\infty$.

Remark 5.7 (Other edges). The edges $t \to \infty$, $s \to \infty$ and $t = 0$ of the (s,t)-square, and the reflection $(s,t) \mapsto (1/s, 1/t)$, can be treated through a conjugation of $e_k \star (\cdot)$ by an operator that is diagonal on Macdonald polynomials; such a conjugation is implicit in [DFK19]. We do not develop this here and use none of it: the only other edge that enters below is $t = 0$, and for it we cite [DFK18] (Remark 6.7).

6. The block valuation law at $s = 1$

6.1. **Taylor pieces.** We write v for the $(s-1)$-adic valuation on $\mathbb{Q}(t)[s]$, $I = (e_1, e_2, \dots)$ for the augmentation ideal (so I^b is spanned by the e_μ with $\ell(\mu) \geq b$), and $\mathrm{lin}_e G := [e_n]G$ for G homogeneous of degree n; lin_e vanishes on I^2. For ρ a partition, a homogeneous symmetric function is *of type* ρ if it lies in $\mathrm{span}\{e_\nu : \nu \trianglerighteq \rho\}$; by Lemmas 4.3 and 4.4 this means $D_j(P) \leq \sum_i \min(\rho_i, j)$ for all j. For a set C of positions, λ_C is the partition formed by the parts λ_i, $i \in C$.

In Theorem 2.5 the variable s enters only through $F|_{x_i \mapsto s x_i \,(i \in A)} = s^{\Delta_A} F$, $\Delta_A = \sum_{i \in A} x_i \partial_i$. Since $s^{\Delta_A} x^\alpha = s^{\alpha \cdot 1_A} x^\alpha = \sum_p (s-1)^p \binom{\Delta_A}{p} x^\alpha$,

$$E_k = \sum_{p \geq 0} (s-1)^p E_k^{(p)}, \qquad E_k^{(p)} = \sum_{|A| = k} c_A x_A \binom{\Delta_A}{p}. \tag{6.1}$$

As $E_k F$ is a symmetric polynomial in x and s (Theorem 4.1), each $E_k^{(p)}$ preserves Λ_m. Since $\binom{\Delta_A}{p}$ is a polynomial of degree p in a derivation, and multiplication by $c_A x_A$ preserves this, $E_k^{(p)}$ is a differential operator of order $\leq p$: all $(p+1)$-fold commutators $[\cdots [E_k^{(p)}, a_0], \dots, a_p]$ with multiplication operators vanish. Moreover $E_k^{(0)}$ is multiplication by e_k (by the proof of Theorem 4.1, $(E_k F)|_{s=1} = e_k F|_{s=1}$), and $E_k^{(p)}(1) = 0$ for $p \geq 1$.

6.2. **First order.**

Theorem 6.1 (First order is a biderivation). *Let $D_k := E_k^{(1)}$ and $B(f,g) := \partial_s (f \star g)|_{s=1}$ for $f, g \in \Lambda_m$ with coefficients in $\mathbb{Q}(t)$ (independent of s). Then D_k is a derivation of Λ_m, and*

$$B(f,g) = \sum_{k,l} M_{kl} \frac{\partial f}{\partial e_k} \frac{\partial g}{\partial e_l}, \qquad M_{kl} := D_k(e_l) = B(e_k, e_l) = M_{lk}.$$

So B is a symmetric biderivation, $B(f,g) = L(fg) - fL(g) - gL(f)$ for $L = \frac{1}{2} \sum M_{kl} \partial^2 / \partial e_k \partial e_l$.

Proof. $D_k = \sum_A c_A x_A \Delta_A$ is a sum of derivations. By Theorem 4.1 the $e_\rho^\star$ form a basis whose transition matrix to the e_ρ is unitriangular at $s = 1$; write $f = \sum_\rho a_\rho(s) e_\rho^\star$ with a_ρ regular at $s = 1$. By associativity, $f \star g = \sum_\rho a_\rho(s) E_{\rho_1} \cdots E_{\rho_\ell} g$. By (6.1),

$$B(f,g) = \sum_\rho a_\rho'(1) e_\rho g + \sum_\rho a_\rho(1) \sum_i \Big(\prod_{j \neq i} e_{\rho_j} \Big) D_{\rho_i} g.$$

Applying the same computation to $f = \sum_\rho a_\rho(s) E_{\rho_1} \cdots E_{\rho_\ell}(1)$ itself, which does not depend on s, and using $D_k(1) = 0$, gives $\sum_\rho a_\rho'(1) e_\rho = 0$. Since $f|_{s=1} = \sum_\rho a_\rho(1) e_\rho$, the remaining sum is $\sum_k \frac{\partial f}{\partial e_k} D_k(g)$, and $D_k(g) = \sum_l \frac{\partial g}{\partial e_l} D_k(e_l)$. Symmetry of M is commutativity of $\star$ [Hik25]. □

Proposition 6.2 (The matrix M). *For $k \geq r \geq 0$, with $L(a,b) := \dfrac{(1 - t^{a+b})(t^{ab} - 1)}{(1 - t^a)(1 - t^b)}$,*

$$M_{kr} = B(e_k, e_r) = r\, e_k e_r + \sum_{j=1}^{r} L(k - r + j,\, j)\, e_{k+j} e_{r-j}.$$

Consequently $[(s-1)^1] e_\lambda^\star = \sum_{i<j} M_{\lambda_i \lambda_j}\, e_{\lambda \setminus \{\lambda_i, \lambda_j\}}$.

Proof. $M_{kr} = [(s-1)^1]\, e_k \star e_r$. Use the form of the Pieri rule in Corollary 8.6 (Section 8; its proof does not use Proposition 6.2): $e_k \star e_r = \sum_{y=0}^{r} s^y \mathcal{C}_{k-y,r-y} e_{k+r-y} e_y$. Since $\pi_a(b)|_{s=1} = \delta_{b0}$, every $\mathcal{C}_{a,b}$ with $b \geq 1$ vanishes at $s = 1$; so the $y = r$ term $s^r e_k e_r$ contributes $r\, e_k e_r$, and the term $y = r - j$, $j \geq 1$, contributes $\partial_s \mathcal{C}_{a,j}|_{s=1}$ with $a = k - r + j$. Now $\partial_s \pi_a(j)|_{s=1} = [u^j] \sum_{m<a} \frac{ut^m}{1+ut^m} = (-1)^{j-1}[a]_{t^j}$, so $\partial_s \mathcal{C}_{a,j}|_{s=1} = -\frac{[a+j]}{[a]}[a]_{t^j} = L(a,j)$. For the second statement, $[(s-1)^1] E_{\lambda_1} \cdots E_{\lambda_\ell}(1) = \sum_i \prod_{j<i} e_{\lambda_j}\, D_{\lambda_i}\big(\prod_{j>i} e_{\lambda_j}\big)$; expand with the Leibniz rule. □

6.3. **Blocks.**

Lemma 6.3 (Polarization)**.** *Let D be a linear operator on a commutative ring and $g_1, \dots, g_r$ elements of it. For $S \subseteq [r]$ let*

$$\delta_S := [\cdots [D, g_{s_1}], \dots, g_{s_q}](1) = \sum_{T \subseteq S} (-1)^{|S \setminus T|} \prod_{i \in S \setminus T} g_i\, D\Big(\prod_{i \in T} g_i\Big).$$

Then

$$D(g_1 \cdots g_r) = \sum_{S \subseteq [r]} \Big(\prod_{i \notin S} g_i\Big) \delta_S,$$

and if D has order $\leq p$, then $\delta_S = 0$ for $|S| > p$, and for $|S| = p$ the map $(g_i)_{i \in S} \mapsto \delta_S$ is a derivation in each argument.

Proof. Substituting the second expression for δ_S, the right side is

$$\sum_{T} \prod_{i \notin T} g_i\, D\Big(\prod_{i \in T} g_i\Big) \sum_{S \supseteq T} (-1)^{|S \setminus T|},$$

and the inner sum is $[T = [r]]$. Vanishing is the definition of order $\leq p$. For $|S| = p$, $D' := [\cdots [D, g_{s_2}], \dots, g_{s_p}]$ has order ≤ 1, so $D'(ab) = aD'(b) + bD'(a) - abD'(1)$, i.e. $[D', ab](1) = a[D', b](1) + b[D', a](1)$; and the iterated commutator is symmetric in its arguments. □

Proposition 6.4 (Block expansion)**.** *Let λ have ℓ parts and $p_1, \dots, p_\ell \geq 0$ with $\sum_i p_i = m$. Then*

$$E_{\lambda_1}^{(p_1)} \cdots E_{\lambda_\ell}^{(p_\ell)}(1) = \sum_{\pi} \prod_{C \in \pi} g_C^{\pi},$$

a finite sum over set partitions π of $[\ell]$ with at least $\ell - m$ blocks, where each g_C^{π} is homogeneous of type λ_C.

Proof. Induction on the number of operators applied, from the right; the empty product 1 has no blocks. Apply $D = E_k^{(p)}$, $k = \lambda_j$, $p = p_j$, to a term $\prod_{C \in \pi} g_C$. By Lemma 6.3 the result is $\sum_{S \subseteq \pi, |S| \leq p} \prod_{C \notin S} g_C \cdot \delta_S$, and we let $\pi' = (\pi \setminus S) \cup \{C'\}$ with $C' = \{j\} \cup \bigcup S$. Then $|\pi'| \geq |\pi| + 1 - p$, which gives the block count. Each δ_S is a combination of products of the g_C and of D applied to such products. Rescaling monomials does not enlarge monomial supports, so the degree step Lemma 4.5 holds for $E_k^{(p)}$ as well, and D_j is additive on products; hence $D_j(\delta_S) \leq \sum_{C \in S} \sum_{i \in C} \min(\lambda_i, j) + \min(\lambda_j, j)$, i.e. δ_S has type $\lambda_{C'}$. □

Definition 6.5. For $\lambda, \mu \vdash n$, $\kappa(\lambda, \mu)$ is the largest r such that $\lambda = \lambda^1 \sqcup \cdots \sqcup \lambda^r$ and $\mu = \mu^1 \sqcup \cdots \sqcup \mu^r$ as multisets of parts, with every block nonempty and $\mu^i \trianglerighteq \lambda^i$ (so $|\mu^i| = |\lambda^i|$); $\kappa = -\infty$ if $\mu \not\trianglerighteq \lambda$. μ is a *coarsening* of λ if $\kappa(\lambda, \mu) = \ell(\mu)$.

6.4. **The block valuation law.**

Theorem 6.6 (Block valuation law)**.** *For all $\lambda, \mu \vdash n$ and all t, $v(c_{\lambda\mu}) \geq \ell(\lambda) - \kappa(\lambda, \mu)$. For $\mu \trianglerighteq \lambda$, $\mu \neq \lambda$, equality holds over $\mathbb{Q}(t)$ and at $t = 0$:*

$$v(c_{\lambda\mu}) = \ell(\lambda) - \kappa(\lambda, \mu).$$

At $t = 0$ the leading coefficient is $(-1)^{\ell-\kappa} N(\lambda, \mu)$ where $N(\lambda, \mu) \geq 1$ is the number of $(k_{ij})_{1 \leq i < j \leq \ell}$ in $\mathbb{Z}_{\geq 0}$ with exactly $\ell - \kappa$ nonzero entries such that $\lambda + \sum k_{ij}(e_i - e_j)$ is a rearrangement of μ (e_i the unit vectors; these are the minimal raising-operator configurations of the proof).

Proof of the lower bound. By (6.1), $[(s-1)^m] e^\star_\lambda = \sum_{\sum p_i = m} E^{(p_1)}_{\lambda_1} \cdots E^{(p_\ell)}_{\lambda_\ell}(1)$. By Proposition 6.4 each term is a sum of products $\prod_{C \in \pi} g_C$ with $|\pi| \geq \ell - m$ and g_C of type λ_C, and such a product lies in $\operatorname{span}\{e_{\sqcup_C \nu^C} : \nu^C \trianglerighteq \lambda_C\}$. So e_μ can occur only if $\kappa(\lambda, \mu) \geq |\pi| \geq \ell - m$. The argument works over $\mathbb{Q}(t)$ and for every specialisation $t \in \mathbb{Q}$. □

Remark 6.7 (Input at $t = 0$). The $t = 0$ edge $e^\star_\mu|_{t=0} = s^{n(\mu)} P_{\mu'}(x; 1/s, 0) = \omega \widetilde{H}_\mu(x; s)$, where $\widetilde{H}_\mu(x; s) = s^{n(\mu)} Q'_\mu(x; s^{-1})$, is **not ours**: it is assembled from [DFK18] eq. (5.15) (with $M_{k,1} = E_k|_{t=0}$), Cor. 5.8 in its level-one form (5.25), Cor. 5.18 (eq. (5.27)) (numbering of arXiv:1505.01657v2), and [Mac95, VI (5.1), (4.14)(iv)]. In particular Theorem 6.6 does not depend on Cherednik's nonsymmetric theory.

Proof of equality. By Remark 6.7, $c_{\lambda\mu}(s, 0) = [h_\mu]\, s^{n(\lambda)} Q'_\lambda(x; s^{-1})$. By [Mac95, III (2.15′) with §5 Ex. 7(a)], $Q'_\lambda(x; u) = \prod_{i<j\leq\ell} \frac{1-R_{ij}}{1-uR_{ij}} h_\lambda$, where $R_{ij} h_\alpha = h_{\alpha + e_i - e_j}$, $h_\alpha = h_{\alpha_1} h_{\alpha_2} \cdots$ and $h_r = 0$ for $r < 0$. Expand $\frac{1-R}{1-uR} = 1 + \sum_{a \geq 1} (u-1) u^{a-1} R^a$; at $u = s^{-1}$ the coefficient of R^a is $-(s-1)s^{-a}$. A *configuration* is a function $\mathbf{m}$ on pairs $i < j \leq \ell$ with values in $\mathbb{Z}_{\geq 0}$; its edge set is $E(\mathbf{m}) = \{ij : \mathbf{m}_{ij} \geq 1\}$ and its result is $\alpha(\mathbf{m}) = \lambda + \sum \mathbf{m}_{ij}(e_i - e_j)$. Then

$$c_{\lambda\mu}(s, 0) = \sum_{\mathbf{m}:\ \alpha(\mathbf{m}) \geq 0,\ \operatorname{sort} \alpha(\mathbf{m}) = \mu} (-1)^{|E(\mathbf{m})|} (s-1)^{|E(\mathbf{m})|} s^{a(\mathbf{m})}, \qquad a(\mathbf{m}) \in \mathbb{Z},$$

a finite sum (the conditions $\alpha_\ell \geq 0$, $\alpha_{\ell-1} \geq 0, \ldots$ bound the $\mathbf{m}_{ij}$ successively). Every configuration contributes $(-1)^{|E|}$ at order $(s-1)^{|E|}$, so there is no cancellation at the lowest order: $v(c_{\lambda\mu}(s, 0)) = E_{\min}$, the least $|E(\mathbf{m})|$ over admissible $\mathbf{m}$, with leading coefficient $(-1)^{E_{\min}} N(\lambda, \mu)$, N the number of minimisers.

$E_{\min} \geq \ell - \kappa$: let C be a connected component of the graph $([\ell], E(\mathbf{m}))$. Transfers stay inside C and move mass from a larger index to a smaller one, and λ is weakly decreasing; so the partial sums of $\alpha|_C$ in index order dominate those of λ_C, and sorting only increases them. Hence the components form a compatible block decomposition, there are at most κ of them, and $|E| \geq \ell - \kappa$.

$E_{\min} \leq \ell - \kappa$: take a decomposition with κ blocks. By maximality each block $(\rho, \nu) = (\lambda_C, \mu^C)$ has no compatible refinement. Let $c_1 < \cdots < c_L$ be the positions of C, pad ν with zeros to length L, and put $B_r = \nu_1 + \cdots + \nu_r$, $P_r = \rho_1 + \cdots + \rho_r$. Then $B_r > P_r$ for $1 \leq r < L$: $\geq$ holds by dominance, and $B_r = P_r$ would split (ρ, ν) into the compatible pieces $(\rho_{\leq r}, \nu_{\leq r})$ and $(\rho_{>r}, \nu_{>r})$, both of positive size. Put $\mathbf{m}_{c_r c_{r+1}} = B_r - P_r \geq 1$. Position c_r then ends with $\rho_r + (B_r - P_r) - (B_{r-1} - P_{r-1}) = \nu_r \geq 0$. This uses $L - 1$ edges per block, $\ell - \kappa$ in all.

Finally, $c_{\lambda\mu} \in \mathbb{Q}[s, t]$ (Theorem 4.1); write $c_{\lambda\mu} = \sum_j (s-1)^j a_j(t)$. The lower bound gives $a_j = 0$ for $j < \ell - \kappa$, and $a_{\ell-\kappa}(0) = (-1)^{\ell-\kappa} N(\lambda, \mu) \neq 0$, so $a_{\ell-\kappa} \neq 0$. □

Remark 6.8 (Novelty). We have not found the block law stated anywhere. Its $t = 0$ case is classical in substance: the raising-operator formula follows from [DLT94, eq. (11)] and Jacobi–Trudi, after which the valuation count above is a short exercise. What is new is the lower bound for all t (from the subset formula alone), the exactness over $\mathbb{Q}(t)$, and the merge weights below.

6.5. **Coarsenings and merge weights.**

Theorem 6.9 (Coarsenings: merge-history expansion)**.** *Let $\mu \neq \lambda$ be a coarsening of λ and $m = \ell(\lambda) - \ell(\mu)$. Then $v(c_{\lambda\mu}) = m$, and $\mathrm{Lead}_{\lambda\mu}(t) := [(s-1)^m]c_{\lambda\mu}$ equals*

$$\mathrm{Lead}_{\lambda\mu}(t) = \sum_{\text{tight histories } H \text{ ending at } \mu} \prod_{\text{merges in } H} W_k(J), \qquad \mathrm{Lead}_{\lambda\mu}(0) = (-1)^m \#\{H\} \neq 0,$$

where a (tight) history processes $\lambda_\ell, \dots, \lambda_1$, each part k either opening a block of size k or merging a set of $p \geq 1$ existing (distinguishable) blocks of sizes $J = (j_1, \dots, j_p)$ into one block of size $k + |J|$ via the order-p piece $E_k^{(p)}$, it is tight if the orders add up to m, it ends at μ if the final block sizes are μ, and $W_k(J) := \mathrm{lin}_e\,[\cdots[E_k^{(p)}, e_{j_1}], \dots, e_{j_p}](1)$.

Proof. Since $\ell(\mu) = \ell - m$, the coefficient of e_μ in $[(s-1)^m]e_\lambda^\star$ can be read modulo $I^{\ell-m+1}$. In Proposition 6.4, a product over $|\pi| > \ell - m$ blocks of positive-degree factors lies in $I^{\ell-m+1}$; so only the terms where every step merges exactly p_j blocks survive, i.e. tight histories. At a tight step, δ_S is a derivation in each argument (Lemma 6.3), so $\delta_S(bc, \dots) \in I^2$ for $b, c \in I$, and modulo I^2, $\delta_S(g_1, \dots, g_p)$ depends only on the linear parts $g_i \equiv \gamma_i e_{n_i}$: it is $\prod_i \gamma_i\, W_k(n_1, \dots, n_p)\, e_n$. An opening step contributes e_k, with $\gamma = 1$. The final product of blocks is $\prod_C \gamma_C e_{|C|}$ modulo $I^{\ell-m+1}$, which gives the formula. By Theorem 6.10 below, $W_k(J)|_{t=0} = (-1)^{|J|}$, so every tight history contributes $(-1)^m$ at $t = 0$. Histories exist: in each block of the coarsening, let the last-processed part merge all the others, opened earlier as singletons. So $\mathrm{Lead}_{\lambda\mu}(0) \neq 0$, and $v(c_{\lambda\mu}) = m$ by the lower bound in Theorem 6.6. □

Let $T_k f := \sum_{|A|=k} c_A x_A f(x_A)$ for $f \in \Lambda_k$, where $f(x_A)$ is f evaluated at the k variables x_a, $a \in A$. (This is unrelated to the Hecke operators of Section 2.)

Theorem 6.10 (Merge weights)**.** *With $n = k + |J|$ and $p = |J|$,*

$$W_k(J) = (-1)^p \frac{[n]_t}{[k]_t} \prod_{j \in J} [k]_{t^j}.$$

Consequently $(-1)^m \mathrm{Lead}_{\lambda\mu}(t) > 0$ for coarsenings and every real $t > 0$.

Lemma 6.11. *For $f \in \Lambda_k$ homogeneous of degree d and $n = k + d$:*

$$\mathrm{lin}_e\, T_k(f) = (-1)^d \frac{[n]_t}{[k]_t} f(1, t, \dots, t^{k-1}).$$

Remark 6.12 (Novelty). Lemma 6.11 is a short corollary of [Mac95, III.7 Ex. 2, III (2.2)] and is probably known.

Lemma 6.13. *For $\ell(\rho) \leq k \leq m$, $T_k(P_\rho(x_1, \dots, x_k; t)) = P_{\rho+1^k}(x_1, \dots, x_m; t)$.*

Proof. By [Mac95, III (2.2)], $P_\lambda = \sum_{w \in S_m/S_m^\lambda} w\big(x^\lambda \prod_{i<j,\, \lambda_i > \lambda_j} \frac{x_i - t x_j}{x_i - x_j}\big)$, S_m^λ the stabiliser of λ. For $\lambda = \rho + 1^k$ the positive entries are the first k; a coset is an image set $A = w([k])$ together with a coset of S_k/S_k^ρ; the pairs with $\lambda_i > \lambda_j$ are those with $i \leq k < j$, giving c_A, and those with $i, j \leq k$, $\rho_i > \rho_j$; and $x^\lambda = x_{[k]} x^\rho_{[k]}$. Grouping by A, the inner sum is [Mac95, III (2.2)] for P_ρ in the variables x_A. □

Lemma 6.14. *For $\lambda \vdash n$ with $\ell(\lambda) = k$, $\mathrm{lin}_e\, P_\lambda(x; t) = (-1)^{n-k} \frac{[n]_t}{[k]_t} P_{\lambda - 1^k}(1, t, \dots, t^{k-1}; t)$.*

Proof. Since $[p_n]e_n = (-1)^{n-1}/n$ and e_ν with $\ell(\nu) \geq 2$ has no p_n term, $\mathrm{lin}_e\, G = (-1)^{n-1}\langle G, p_n\rangle = (-1)^{n-1}(1 - t^n)\langle G, p_n\rangle_t$. By $\langle P_\lambda, Q_\mu\rangle_t = \delta_{\lambda\mu}$ and $Q_\lambda = b_\lambda P_\lambda$ [Mac95, III (4.9)], $\langle P_\lambda, p_n\rangle_t = [P_\lambda]p_n / b_\lambda(t)$, and $[P_\lambda]p_n = t^{n(\lambda)}\varphi_{k-1}(t^{-1}) = (-1)^{k-1} t^{n(\lambda) - \binom{k}{2}} \varphi_{k-1}(t)$ [Mac95, III.7 Ex. 2]. On

the other side, with $\rho = \lambda - 1^k$, [Mac95, III.2 Ex. 1] gives $P_\rho(1,\dots,t^{k-1}) = t^{n(\rho)}\varphi_k / \prod_{i\ge 0}\varphi_{m_i(\rho)}$ with $m_0(\rho) = k - \ell(\rho)$; here $m_i(\rho) = m_{i+1}(\lambda)$, so the denominator is b_λ, and $n(\rho) = n(\lambda) - \binom{k}{2}$. Since $\frac{[n]}{[k]}\varphi_k = (1-t^n)\varphi_{k-1}$, the two sides agree. □

Proof of Lemma 6.11. The $P_\rho(x_1,\dots,x_k;t)$ with $\ell(\rho)\le k$ form a basis of Λ_k [Mac95, III (2.7)]; for $f = P_\rho$, $\rho \vdash d$, combine Lemmas 6.13 and 6.14. □

Proof of Theorem 6.10. Let $J = (j_1,\dots,j_p)$, $d = |J|$. First, $W_k(J) = \operatorname{lin}_e E_k^{(p)}(e_{j_1}\cdots e_{j_p})$: in Lemma 6.3, $\delta_\varnothing = E_k^{(p)}(1) = 0$, and for $\varnothing \ne S \ne [p]$ the term $\prod_{i\notin S} e_{j_i}\,\delta_S$ lies in I^2. Next, with $E(u) = \sum_r e_r u^r$ and $R(u) := \prod_{a\in A}\frac{1+sux_a}{1+ux_a}$ read as a function of x_A,

$$E_k\Big(\prod_{i=1}^{p} E(u_i)\Big) = \prod_i E(u_i)\cdot T_k\Big(\prod_i R(u_i)\Big), \qquad \log R(u) = \sum_{r\ge 1}(-1)^{r-1}\frac{s^r-1}{r}u^r p_r(x_A).$$

Take $[u_1^{j_1}\cdots u_p^{j_p}]$ and $[(s-1)^p]$. Terms in which some e_{r_i}, $r_i \ge 1$, is taken from $\prod E(u_i)$ lie in I^2. In the remaining term, each $[u_i^{j_i}]R(u_i)$ has $(s-1)$-valuation ≥ 1, with $(s-1)^1$-coefficient $(-1)^{j_i-1}p_{j_i}(x_A)$. Hence $W_k(J) = (-1)^{d-p}\operatorname{lin}_e T_k(p_J)$, and Lemma 6.11 with $p_j(1,\dots,t^{k-1}) = [k]_{t^j}$ gives the formula. For $t > 0$ every $W_k(J)$ has sign $(-1)^{|J|}$, and the orders along a tight history add up to m. □

7. Leading coefficients: graph cumulants and blocks

Throughout this section μ is a coarsening of λ, $\ell = \ell(\lambda)$, $m = \ell - \ell(\mu)$, and the parts of λ are indexed by positions $[\ell]$. Everything rests on Theorems 6.9 and 6.10.

Lemma 7.1 (Histories are increasing forests)**.** *Tight histories of λ are in bijection with increasing forests on $[\ell]$ (every child larger than its parent): when position i merges existing blocks, i becomes the parent of their roots. The trees are the final blocks, and the orders automatically add up to $\ell - \#\text{trees}$.*

Proof. Every existing block at step i was built from positions $> i$, so all edges increase. Conversely, an increasing forest determines the history: at step i, merge the subtrees rooted at the children of i, which are complete by then. Each edge accounts for one unit of order. □

Lemma 7.2 (Telescoping)**.** *For an increasing tree T on a set C of positions, let N_v be the λ-mass of the subtree at v. The product of the merge weights along the history of T is*

$$\prod_v W_{\lambda_v}\big((N_c)_{c\ \text{child of}\ v}\big) = \frac{1-t^{|\lambda_C|}}{\prod_{i\in C}(1-t^{\lambda_i})}\prod_{\text{edges } v\to c}(t^{\lambda_v N_c}-1).$$

Proof. By Theorem 6.10, the merge at v contributes $(-1)^{\#\text{children}}\frac{1-t^{N_v}}{1-t^{\lambda_v}}\prod_c \frac{1-t^{\lambda_v N_c}}{1-t^{N_c}}$ (a leaf contributes 1). Every non-root c is a child exactly once, so its factor $1-t^{N_c}$ cancels against its own numerator, leaving the root mass $|\lambda_C|$; and $(-1)^{\#\text{edges}}\prod(1-x) = \prod(x-1)$. □

Lemma 7.3 (Tree–graph identity)**.** *For indeterminates $x_{ij} = x_{ji}$ $(i \ne j \in [\ell])$,*

$$\sum_{\text{increasing trees } T \text{ on } [\ell]}\ \prod_{\text{edges } v\to c}\Big(\prod_{j\in T_c} x_{vj} - 1\Big) = \sum_{\text{connected graphs } H \text{ on } [\ell]}\ \prod_{e\in H}(x_e - 1),$$

where T_c is the vertex set of the subtree at c.

Proof. Map a connected graph H to an increasing tree recursively: the root is $\min V$, the children are the minima of the components of $H - \min V$, and recurse on each component. Fix T and sum over the H mapping to it. At each vertex v and child c, the edges from v into T_c form an arbitrary nonempty set $S \subseteq T_c$ (nonempty since T_c must be attached to v; otherwise free, since they do not change the components inside T_c); there are no edges between different child sets; and the edges inside T_c are accounted for recursively. So the fibre sum factors, and $\sum_{\varnothing \neq S \subseteq T_c} \prod_{j \in S}(x_{vj} - 1) = \prod_{j \in T_c} x_{vj} - 1$. □

Theorem 7.4 (Full-merge lead)**.** *For $\lambda \vdash n$ with $\ell = \ell(\lambda) \geq 2$,*

$$\mathrm{Lead}_{\lambda,(n)}(t) = \frac{1 - t^n}{\prod_i (1 - t^{\lambda_i})} K_\lambda(t), \qquad K_\lambda(t) := \sum_{H \text{ connected on } [\ell]} \prod_{ij \in H} (t^{\lambda_i \lambda_j} - 1).$$

K_λ is the connected part (joint cumulant) of $t^{e_2(\lambda)} = \prod_{i<j}(1 + (t^{\lambda_i \lambda_j} - 1))$.

Proof. By Theorem 6.9 with $\mu = (n)$ and Lemma 7.1, $\mathrm{Lead}_{\lambda,(n)}$ is a sum over increasing trees on $[\ell]$; by Lemma 7.2 each tree contributes $\frac{1-t^n}{\prod_i (1-t^{\lambda_i})} \prod_{v \to c}(t^{\lambda_v N_c} - 1)$; and Lemma 7.3 with $x_{ij} = t^{\lambda_i \lambda_j}$, for which $\prod_{j \in T_c} x_{vj} = t^{\lambda_v N_c}$, converts the tree sum into K_λ. The cumulant statement is the exponential formula. □

Corollary 7.5. *(a) $\lambda = 1^n$: $\mathrm{Lead} = (-1)^{n-1}[n]_t\, I_n(t)$ with I_n the tree inversion enumerator [MR68]. (b) $t = 0$: $\mathrm{Lead} = (-1)^{\ell-1}(\ell - 1)!$. (c) $t \to 1$: $\mathrm{Lead} \to (-1)^{\ell-1} n^{\ell-1}$ (weighted Cayley). (d) $\operatorname{sgn} \mathrm{Lead} = (-1)^{\ell-1}$ for all $t > 0$.*

Proof. (a) For $\lambda = 1^n$ the prefactor is $[n]_t/(1-t)^{n-1}$ and $K_\lambda = \sum_{H \text{ conn.}} (t-1)^{|H|} = (t-1)^{n-1} I_n(t)$ [MR68]. (b) At $t = 0$ the prefactor is 1, and in the tree form every edge factor is -1; there are $(\ell - 1)!$ increasing trees. (c) Put $\varepsilon = t - 1$. In K_λ only spanning trees contribute to the lowest order $\varepsilon^{\ell-1}$, with $t^{\lambda_i \lambda_j} - 1 = \lambda_i \lambda_j \varepsilon + O(\varepsilon^2)$, and $\sum_T \prod_{ij \in T} \lambda_i \lambda_j = \sum_T \prod_i \lambda_i^{\deg_T(i)} = (\prod_i \lambda_i)\, n^{\ell-2}$ by the vertex-weighted Cayley formula (each i occurs $\deg_T(i) - 1$ times in the Prüfer code). The prefactor is $(-n\varepsilon)/((-\varepsilon)^\ell \prod_i \lambda_i) \cdot (1 + O(\varepsilon))$. (d) is the case $\mu = (n)$, $m = \ell - 1$, of the sign statement in Theorem 6.10: each merge contributes $(-1)^p$ times a positive quantity, $\sum p = m$ along every history, and histories exist. □

Remark 7.6 (Prior art — graph half). The tree, connected-graph and cumulant forms of K_λ are a special case of Dołęga [Doł19, Prop. 2.1 and (11), Lemma 2.3] ($K_\lambda = (t-1)^{\ell-1} T(1,t)$ for the Tutte polynomial of the multigraph with $\lambda_i \lambda_j$ edges between i and j), cf. also [GS96; MR68]; step (3) is a Penrose-type tree–graph identity [Pen67]. Ours: the identification of the $\star$-lead with prefactor$\cdot K_\lambda$ (Theorems 6.9, 6.10 and 7.4) and Theorems 7.7 and 7.8. Haglund–Tewari's single-row cumulant [HT26, Def. 7.1, Thm. 7.3] (their q is our t) is the same graph half: since $\langle \widetilde{H}_{(m)}, e_m \rangle = q^{\binom{m}{2}}$, their cumulant $\kappa(\lambda)$ (we write C_λ) satisfies $\langle C_\lambda, e_n \rangle = t^{n(\lambda')}(t-1)^{1-\ell} K_\lambda(t)$. *Separator.* Full-merge leads $L(\lambda)$ of any such theory are only defined up to the gauge $L(\lambda) \mapsto L(\lambda)\, a^{\ell(\lambda)} g(|\lambda|) \prod_i f(\lambda_i)$ (rescaling generators, degree-wise normalisation, rescaling the expansion parameter). The first gauge invariant appears at $n = 4$: $J := L(1^4)L(2,2)/L(2,1,1)^2$ (the exponents of a, $g(4)$, $f(1)$ and $f(2)$ all cancel). For $\star$, Theorem 7.4 gives $J = \frac{(t^2+1)(t^3+3t^2+6t+6)}{(t+1)(t^2+t+2)^2}$, and Haglund–Tewari's $\langle C_\lambda, e_n \rangle$ gives the same value. Dołęga's $\oplus$-cumulant of columns (lowest $(q-1)$-coefficient of $[g_n]\widetilde{H}_{\lambda'}$, $g_k = \widetilde{H}_{1^k}$) gives, by computation, $J \equiv 3/2 = J_\star(0)$, whereas $J_\star(1) = 1$, the Cayley value. So up to gauge Dołęga's $\oplus$ is t-independent and agrees with $\star$ only at $t = 0$, where the leads are partition-lattice Möbius values (Corollary 7.5(b)), while Haglund–Tewari's is the same graph object without $\star$.

Theorem 7.7 (Coarsening leads factor)**.** *For μ a coarsening of λ,*

$$\mathrm{Lead}_{\lambda\mu} = \sum_{\pi} \prod_{B\in\pi} \mathrm{Lead}_{\lambda_B,(|\lambda_B|)},$$

summed over set partitions π of the positions $[\ell]$ whose sorted block sums equal μ (singletons contribute 1). There is no interleaving factor.

Proof. By Lemma 7.1, tight histories ending at μ are increasing forests whose trees have sorted masses μ, i.e. a set partition π with block sums μ together with an increasing tree on each block. By Lemma 7.2 the weight is a product over the trees, each depending only on its own parts; summing over the trees on each block gives $\mathrm{Lead}_{\lambda_B,(|\lambda_B|)}$ by Theorem 7.4 (and 1 for a singleton). □

Theorem 7.8 (Block multiplicativity)**.** *Let $\kappa = \kappa(\lambda,\mu) \geq 1$, $\ell = \ell(\lambda)$, $m = \ell - \kappa$. Then*

$$[(s-1)^m]\, c_{\lambda\mu} = \sum_{\pi} \sum_{(\nu^C)_{C\in\pi}} \prod_{C\in\pi} [(s-1)^{|C|-1}]\, c_{\lambda_C,\nu^C},$$

where π runs over set partitions of $[\ell]$ into exactly κ blocks and (ν^C) over tuples with $\bigsqcup_C \nu^C = \mu$ and $\nu^C \trianglerighteq \lambda_C$; every factor then has $\kappa(\lambda_C,\nu^C) = 1$. With Theorem 6.6 this is $\mathrm{Lead}_{\lambda\mu}$.

Proof. Expand $[(s-1)^m]e^\star_\lambda = \sum_{\sum p_i = m} E^{(p_1)}_{\lambda_1} \cdots E^{(p_\ell)}_{\lambda_\ell}(1)$ step by step with Lemma 6.3, as in Proposition 6.4: this is a sum over histories H (step i chooses a set S_i of current blocks with $|S_i| \leq p_i$), each giving $\prod_{C\in\pi(H)} g^H_C$ with g^H_C of type λ_C, and $|\pi(H)| = \ell - \sum |S_i| \geq \ell - m = \kappa$, with equality iff $|S_i| = p_i$ for all i (tight). A non-tight term has more than κ blocks, so its e_μ-coefficient vanishes by maximality of κ. In a tight history, step i only acts, through δ_{S_i}, on the blocks it merges; so g^H_C is the function produced by the restricted history on C, which is a single-block tight history of the composition λ_C (positions of C in their order), since $\sum_{i\in C} |S_i| = |C|-1$. Conversely, single-block tight histories on the blocks of π interleave uniquely, by position, into a global tight history. Hence the tight part is $\sum_{|\pi|=\kappa} \prod_{C\in\pi} \mathrm{Conn}(\lambda_C)$, where $\mathrm{Conn}(\lambda_C)$ is the sum over single-block tight histories of λ_C. Applying the same expansion to λ_C with $m = |C| - 1$ shows $[e_\nu]\mathrm{Conn}(\lambda_C) = [e_\nu][(s-1)^{|C|-1}]e^\star_{\lambda_C}$ whenever $\kappa(\lambda_C,\nu) = 1$. Finally, $[e_\mu]\prod_C \mathrm{Conn}(\lambda_C) = \sum_{(\nu^C)} \prod_C [e_{\nu^C}]\mathrm{Conn}(\lambda_C)$ over tuples with $\bigsqcup_C \nu^C = \mu$; factors with $\nu^C \not\trianglerighteq \lambda_C$ vanish by type, and $\kappa(\lambda_C,\nu^C) \geq 2$ would refine π to more than κ compatible blocks. Commutativity of $\star$ [Hik25] identifies the coefficients for the composition λ_C with those for the sorted partition. □

Remark 7.9. Theorem 7.7 is the case where each ν^C is a single part. The factorisation has the shape of a Mayer (cluster) expansion, i.e. moment–cumulant inversion of the graph identity in Theorem 7.4; the $\star$-specific content is that the connected leads are those of Theorem 7.4, with no interleaving correction.

8. Box Complement and the linear coefficient

In this section zero parts are allowed: λ is padded with zeros to a prescribed length, $E_0 = \mathrm{id}$, and $c_{\lambda\mu}$ refers to the nonzero parts. By Lemma 2.8, in N variables (for every N, including N below the degree) $e^\star_\lambda(x_1,\dots,x_N) = \sum_\mu c_{\lambda\mu} e_\mu(x_1,\dots,x_N)$, where $e_\mu = 0$ if $\mu_1 > N$.

Theorem 8.1 (Box Complement)**.** *Let λ,μ have at most ℓ parts (padded with zeros) and all parts $\leq N$; put $N^\ell - \lambda := (N-\lambda_\ell,\dots,N-\lambda_1)$. Then*

$$c_{N^\ell-\lambda,\,N^\ell-\mu}(s,t) = s^{N\binom{\ell}{2}-(\ell-1)|\lambda|}\, c_{\lambda\mu}(s,t).$$

Lemma 8.2 (Inversion)**.** *In N variables, let G be homogeneous of degree g and $F(x) := e_N(x)^m\, G(1/x)$. Then*

$$E_{N-k}F = s^{m(N-k)-g}\, e_N^{m+1}\, (E_kG)(1/x).$$

Proof. Write $y = 1/x$ and index the terms of E_{N-k} by the complement A ($|A| = k$) of the $(N-k)$-subset B. Then $c_B(x) = \prod_{i\in B, j\in A} \frac{y_j - t y_i}{y_j - y_i} = c_A(y)$; $x_B = e_N(x)\, y_A$; $F|_{x_i \mapsto s x_i\,(i\in B)} = s^{m(N-k)} e_N(x)^m\, G|_{y_i\mapsto s^{-1}y_i\,(i\in B)}$; and by homogeneity $G|_{y_i\mapsto s^{-1}y_i\,(i\in B)} = s^{-g}\, G|_{y_a\mapsto s y_a\,(a\in A)}$. Summing over A gives $s^{m(N-k)-g} e_N^{m+1} \sum_A c_A(y) y_A G|_{y_a\mapsto s y_a}$. □

Proof of Theorem 8.1. Let $G_j := E_{\lambda_{\ell-j+1}} \cdots E_{\lambda_\ell}(1)$, of degree $D_j = \lambda_\ell + \cdots + \lambda_{\ell-j+1}$, and $F_j := E_{N-\lambda_{\ell-j+1}} \cdots E_{N-\lambda_\ell}(1)$, in N variables. By Lemma 8.2 and induction, $F_j = s^{\sigma_j} e_N^j G_j(1/x)$ with $\sigma_0 = 0$ and $\sigma_{j+1} = \sigma_j + j(N - \lambda_{\ell-j}) - D_j$. Writing $p_i = \lambda_{\ell+1-i}$,

$$\sigma_\ell = N\binom{\ell}{2} - \sum_i (i-1)p_i - \sum_i (\ell - i)p_i = N\binom{\ell}{2} - (\ell-1)|\lambda|.$$

By commutativity of $\star$, $F_\ell = e^\star_{N^\ell - \lambda}$. Since $e_{N-r}(x) = e_N(x) e_r(1/x)$ in N variables, $e_N^\ell e_\mu(1/x) = e_{N^\ell - \mu}(x)$ for $\ell(\mu) \le \ell$, and every μ in the support of $e^\star_\lambda$ has $\ell(\mu) \le \ell(\lambda) \le \ell$ (Theorem 4.1). Hence $e^\star_{N^\ell-\lambda} = s^{\sigma_\ell} \sum_{\mu_1 \le N} c_{\lambda\mu}\, e_{N^\ell - \mu}(x)$. The map $\mu \mapsto N^\ell - \mu$ is a bijection between partitions with at most ℓ parts, all $\le N$; every ν in the support of $e^\star_{N^\ell-\lambda}$ has at most ℓ parts; and the e_ν with $\nu_1 \le N$ form a basis of Λ_N. Compare coefficients. □

Corollary 8.3 (Column Lemma; lead invariance)**.** $c_{\lambda + 1^\ell, \mu + 1^\ell} = s^{\binom{\ell}{2}} c_{\lambda\mu}$. *Hence v and* Lead *are invariant under $(\lambda, \mu) \mapsto (\lambda + 1^\ell, \mu + 1^\ell)$ and $(\lambda, \mu) \mapsto (N^\ell - \lambda, N^\ell - \mu)$; e.g. $(2,2,2) \to (3,3)$ and $(2,2,2) \to (4,1,1)$ have the lead $[3]_t(t+2)$ of $(1,1,1) \to (3)$.*

Proof. Complement with N and then with $N+1$: $(N+1)^\ell - (N^\ell - \lambda) = \lambda + 1^\ell$, and the exponents add up to $\big(N\binom{\ell}{2} - (\ell-1)|\lambda|\big) + \big((N+1)\binom{\ell}{2} - (\ell-1)(N\ell - |\lambda|)\big) = \binom{\ell}{2}$. Since s^σ is a unit of $\mathbb{Q}[[s-1]]$ with constant term 1, valuations and leading coefficients are preserved. For the example, $(2,2,2) \to (4,1,1)$ is $(1,1,1) \to (3,0,0)$ plus a column, and $(2,2,2) \to (3,3)$ is the complement with $N = 3$ of $(1,1,1) \to (3,0,0)$; the lead of $(1,1,1) \to (3)$ is $[3]_t(t+2)$ by Theorem 7.4. □

Remark 8.4 (Prior art). The inversion lemma follows in a few lines from [DFK19, Rem. 3.3] ($S\Gamma_I S = \Gamma_I^{-1}$ under $x \mapsto x^{-1}$) plus subset complement; stability is [Hik25, Prop. 3.8, Cor. 3.10]. We have not found Theorem 8.1 or Corollary 8.3 for $\star$ in the literature.

Theorem 8.5 (Plethystic linear coefficient)**.** *For G homogeneous of degree d: $\mathrm{lin}_e(e_k \star G) = (-1)^d \frac{[k+d]_t}{[k]_t}\, G[(s-1)[k]_t]$, where $G \mapsto G[(s-1)[k]_t]$ is the ring map $p_r \mapsto (s^r - 1)[k]_{t^r}$.*

Proof. By linearity take $G = e_{j_1} \cdots e_{j_p}$, $d = \sum j_i$. As in the proof of Theorem 6.10, $E_k(e_J) = \sum_{w \le J} e_{J-w}\, T_k\big([u^w] \prod_i R(u_i)\big)$. For $w \ne J$ the term is a product of an element of I and a symmetric function of positive degree, hence lies in I^2. For $w = J$, Lemma 6.11 applied to the symmetric function $f = \prod_i [u_i^{j_i}] R(u_i)$ of x_A gives $(-1)^d \frac{[k+d]}{[k]} f(1, t, \ldots, t^{k-1})$, and $[u^j]R(u)$ at $x_A = (1, t, \ldots, t^{k-1})$ is $\pi_k(j) := [u^j] \prod_{m=0}^{k-1} \frac{1+sut^m}{1+ut^m}$. Taking logarithms, $e_j \mapsto \pi_k(j)$ is the ring map $p_r \mapsto (s^r - 1)[k]_{t^r}$. □

Corollary 8.6 (Pieri rule, all s)**.** $e_k \star e_r = \sum_{y=0}^{\min(k,r)} s^y \mathcal{C}_{k-y, r-y}\, e_{k+r-y} e_y$ *with* $\mathcal{C}_{a,0} = \mathcal{C}_{0,b} = 1$, $\mathcal{C}_{a,b} = (-1)^b \frac{[a+b]}{[a]} \pi_a(b)$, $\pi_a(j) = [u^j] \prod_{m=0}^{a-1} \frac{1+sut^m}{1+ut^m}$.

Proof. By Theorem 4.1 and commutativity, the support of $e_k \star e_r$ consists of $e_{k+r-y}e_y$ with $y \le \min(k,r)$. Applying Corollary 8.3 with $\ell = 2$ y times, $c_{(k,r),(k+r-y,y)} = s^y\, c_{(k-y,r-y),(k+r-2y,0)} = s^y \operatorname{lin}_e(e_{k-y} \star e_{r-y})$. If $k - y = 0$ this is s^y; otherwise Theorem 8.5 with $G = e_{r-y}$ gives $s^y(-1)^{r-y}\frac{[k+r-2y]}{[k-y]}\pi_{k-y}(r-y) = s^y\mathcal{C}_{k-y,r-y}$. □

Remark 8.7. This proof of the Pieri rule is independent of the one in Section 3: it uses neither the parabolic kernel computation of Proposition 2.2 beyond the subset formula, nor the recursion of Theorem 3.2. For $r \ge k$ the monomials $e_b e_{r+k-b}$, $0 \le b \le k$, are distinct, so comparing with Corollary 3.3 gives $F_n(t^m) = (-1)^m\frac{[n+m]}{[n]}\pi_n(m)$ for $1 \le n \le m$. Hikita's [Hik25, Thm. 3.12] is the case $k = 1$.

9. Second order: every valuation-2 lead

By Theorem 7.8, valuation-2 leads reduce to connected leads with $\ell(\lambda) = 3$, $\kappa = 1$; by Corollary 8.3 to two-part μ. With T_a as in Lemma 6.11, put

$$\Gamma_a(e_b, e_c) := E_a^{(2)}(e_b e_c) - e_b E_a^{(2)}(e_c) - e_c E_a^{(2)}(e_b) = \sum_{1\le r\le b,\ 1\le q\le c} (-1)^{r+q} e_{b-r} e_{c-q}\, T_a(p_r p_q).$$

Indeed, with $R(u)$ as in the proof of Theorem 6.10, $R(u) = \prod_{x\in A}(1+(s-1)y_x(u))$ with $y_x(u) = \frac{ux_x}{1+ux_x}$, so $[(s-1)^2]R(u)R(v) = e_2(Y_u)+e_2(Y_v)+p_1(Y_u)p_1(Y_v)$ in the alphabets $Y_u = \{y_x(u)\}_{x\in A}$, Y_v; hence $E_a^{(2)}(E(u)E(v)) - E(u)E_a^{(2)}E(v) - E(v)E_a^{(2)}E(u) = E(u)E(v)\,T_a(p_1(Y_u)p_1(Y_v))$, and $p_1(Y_u) = \sum_{r\ge1}(-1)^{r-1}u^r p_r(x_A)$. Take $[u^b v^c]$.

Proposition 9.1. *If (a,b,c) is any ordering of λ and $\kappa(\lambda,\mu) = 1$, then $[(s-1)^2]c_{\lambda\mu} = [e_\mu](\Gamma_a(e_b,e_c) + D_aD_b(e_c))$.*

Proof. As $\star$ is commutative and associative (Section 2), $e_\lambda^\star = E_aE_bE_c(1)$ for every ordering (a,b,c) of λ; fix one. Since $E^{(p)}(1) = 0$ for $p \ge 1$, $[(s-1)^2]E_aE_bE_c(1) = E_a^{(2)}(e_be_c) + D_aD_b(e_c) + e_aE_b^{(2)}(e_c)$, and $E_a^{(2)}(e_be_c) = \Gamma_a(e_b,e_c)+e_bE_a^{(2)}(e_c)+e_cE_a^{(2)}(e_b)$. So the surviving expansion terms are $E_a^{(2)}(e_be_c)$, $e_aE_b^{(2)}(e_c)$ and $D_aD_b(e_c)$, while the three terms to discard, $e_aE_b^{(2)}(e_c)$, $e_bE_a^{(2)}(e_c)$, $e_cE_a^{(2)}(e_b)$, are not these three: they are one term discarded from the expansion plus the two introduced by the Γ_a packaging (the two triples share one element), and this list is complete. Each of them has support in $\kappa(\lambda,\mu) \ge 2$. Indeed, $E_c(1) = e_c$, so $E_a(e_c) = E_aE_c(1) = e^\star_{(a,c)}$ (by commutativity) and $E_a^{(2)}(e_c) = [(s-1)^2]\,e^\star_{(a,c)}$. By Theorem 4.1, $e^\star_{(a,c)}$ is supported on $\{e_\nu : \nu \trianglerighteq (a,c)\}$, hence so is $E_a^{(2)}(e_c)$, and $e_bE_a^{(2)}(e_c)$ is supported on $\{e_{\nu\sqcup(b)}\}$. For such $\mu = \nu \sqcup (b)$, the blocks $\lambda^1 = \mu^1 = \{b\}$, $\lambda^2 = \{a,c\}$, $\mu^2 = \nu$ form a two-block decomposition, so $\kappa(\lambda,\mu) \ge 2$. The other two terms are identical with the letters permuted. □

Theorem 9.2 (Constant-term adjoint formula)**.** *For $g \in \Lambda_a$ and $F \in \Lambda$,*

$$\varphi_a(t)\,\langle T_a g, F\rangle_t = \mathrm{CT}\big[Z\, g(z)\, F(z_1^{-1},\dots,z_a^{-1})\,\Omega(z)\big],$$

where $Z = z_1\cdots z_a$ and $\Omega(z) = \prod_{1\le i<j\le a}\frac{z_j-z_i}{z_j-tz_i}$, expanded in z_i/z_j $(i<j)$.

Proof. Let $G = T_ag$. In exactly a variables only $A = [a]$ survives and $c_{[a]} = 1$, so $G(z) = Zg(z)$; and $G \in \operatorname{span}\{P_\mu : \ell(\mu) = a\}$ by Lemma 6.13. We read CT of a Laurent polynomial times Ω either formally or as the integral over the torus $|z_i| = 1$ for $|t| < 1$; the two agree because the expansion of Ω converges there, and both sides of the theorem are rational in t.

Step 1. For $\alpha \in \mathbb{Z}^a_{\geq 0}$, $\langle G, q_\alpha\rangle_t = [y^\alpha]G(y_1,\dots,y_a)$, with $q_\alpha = \prod q_{\alpha_i}$: pair G with the Cauchy identity $\sum_\mu Q_\mu(x)P_\mu(y) = \prod_{i,j}\frac{1-tx_iy_j}{1-x_iy_j} = \sum_\alpha q_\alpha(x)y^\alpha$ [Mac95, III (4.2), (4.4)], using $\langle P_\lambda, Q_\mu\rangle_t = \delta_{\lambda\mu}$ [Mac95, III (4.9)].

Step 2. For $\ell(\lambda) \leq a$ (padded), the raising-operator formula [Mac95, III (2.15′)] gives $Q_\lambda = \sum_\beta c_\beta q_{\lambda+\beta}$ with $\sum_\beta c_\beta y^{-\beta} = \prod_{i<j}\frac{1-y_j/y_i}{1-ty_j/y_i}$; operators involving positions beyond $\ell(\lambda)$ only produce negative entries, where q and $[y^{\cdot}]$ both vanish. With step 1 and the relabelling $z_k = y_{a+1-k}$,

$$\langle G, Q_\lambda\rangle_t = \mathrm{CT}\big[G(z)\, z^{-\lambda^{\mathrm{rev}}}\,\Omega(z)\big], \qquad \lambda^{\mathrm{rev}} = (\lambda_a,\dots,\lambda_1).$$

Step 3. Let $\Delta = \prod_{i\neq j}\frac{1-z_i/z_j}{1-tz_i/z_j}$ and $\theta = \prod_{i<j}\frac{z_i-tz_j}{z_i-z_j}$, so that $\Delta\theta = \Omega$. For symmetric f, $f\Omega$ is analytic on the torus and $f\Delta$ is symmetric, so averaging over S_a and using $\sum_w w(\theta) = v_a(t) = [a]_t!$ [Mac95, III (1.4)] gives $\mathrm{CT}[f\Omega] = \frac{v_a}{a!}\mathrm{CT}[f\Delta]$.

Step 4. By [Mac95, III (2.1)–(2.2)], $Q_\lambda(z^{-1}) = \frac{b_\lambda}{v_\lambda}\sum_w w(z^{-\lambda}\tilde\theta)$ with $\tilde\theta = \prod_{i<j}\frac{z_j-tz_i}{z_j-z_i}$ and $v_\lambda = \prod_{i\geq 0} v_{m_i(\lambda)}$, $m_0 = a-\ell(\lambda)$. Since $\tilde\theta\Delta = \prod_{i<j}\frac{z_i-z_j}{z_i-tz_j}$, step 3, symmetry, and the relabelling $z_k \leftrightarrow z_{a+1-k}$ give

$$\mathrm{CT}\big[G\,Q_\lambda(z^{-1})\,\Omega\big] = v_a\frac{b_\lambda}{v_\lambda}\,\mathrm{CT}\big[G\,z^{-\lambda^{\mathrm{rev}}}\Omega\big] = v_a\frac{b_\lambda}{v_\lambda}\langle G, Q_\lambda\rangle_t.$$

Step 5. Write $F = \sum_\lambda F_\lambda Q_\lambda$. Terms with $\ell(\lambda) > a$ vanish on both sides, and $\langle G, Q_\lambda\rangle_t = 0$ unless $\ell(\lambda) = a$, in which case $m_0 = 0$ and $v_a\, b_\lambda/v_\lambda = v_a(1-t)^a = \varphi_a$. Hence $\mathrm{CT}[GF(z^{-1})\Omega] = \varphi_a\langle G, F\rangle_t$. □

Theorem 9.3 (Two-point formula)**.** *Let $g \in \Lambda_a$ be homogeneous of degree d, $n = a+d$, $x, y \geq 1$, $x + y = n$, and $\Phi_a(g; x, y) := \langle T_a g, p_x p_y\rangle$ (Hall pairing, not the $\langle\cdot,\cdot\rangle_t$ of Theorem 9.2). For $1 \leq A \leq a-1$ put $B = a - A$ and $G_A(w) = \sum_k G_{A,k}w^k := g(1, t, \dots, t^{A-1}, w, wt, \dots, wt^{B-1})$. Then*

$$\Phi_a(g;x,y) = (-1)^a(1-t^x)(1-t^y)\Bigg\{\sum_{A=1}^{a-1} t^{-AB}\Big[\frac{G_{A,y-B}}{(1-t^A)(1-t^B)} + \frac{1}{1-t^a}\sum_{m\geq 1}(t^{-Am}-t^{Bm})\,G_{A,y-B-m}\Big] - \frac{g(1,t,\dots,t^{a-1})}{1-t^a}\sum_{j=0}^{a-1} t^{-jy}\Bigg\}.$$

Proof. Let $S(u) := \sum_{i=1}^a \frac{1}{uz_i-1} = \sum_{x\geq 1} u^{-x}p_x(z^{-1})$. By Theorem 9.2 and homogeneity,

$$\mathcal{R}(u,v) := \frac{1}{\varphi_a}\mathrm{CT}\big[g\,Z\,S(u)S(v)\,\Omega\big] = \sum_{x+y=n}\frac{\Phi_a(g;x,y)}{(1-t^x)(1-t^y)}u^{-x}v^{-y} = u^{-n}\rho(w), \qquad w = u/v,$$

with ρ a polynomial, so $\Phi_a(g;x,y) = (1-t^x)(1-t^y)[w^y]\rho$.

String expansion. Take $|t|$ small, u, v generic, and radii $\max(|u|^{-1}, |v|^{-1}) < r_1 < \cdots < r_a$; the constant term is the integral of $g\,S(u)S(v)\,\Omega\prod_m dz_m/(2\pi i)$ over $|z_m| = r_m$. Expand $S(u)S(v)$ into its a^2 summands and integrate $z_1,\dots,z_a$ in turn. When z_m is integrated, its poles inside $|z_m| = r_m$ are at $1/u$ or $1/v$ if z_m carries the u- or v-factor, and at tz_l for $l < m$ (from the factor $\frac{z_m-z_l}{z_m-tz_l}$); the poles z_j/t, $j > m$, lie outside, and g is a polynomial. The numerator $\prod_{l<m}(z_m - z_l)$ kills the residue at tz_l if some earlier $z_{l'}$ already equals tz_l. Hence the evaluation points form *t-strings* $q, qt, \dots, qt^{A-1}$ (at increasing positions), each started by a *seed* at $1/u$ or $1/v$, and position 1 is always a seed. The surviving configurations are: (i) a word $\omega \in \{U, V\}^a$ containing

both letters, the u-factor at the first U and the v-factor at the first V, the $(\alpha+1)$-st U at t^α/u and the $(\beta+1)$-st V at t^β/v; (ii) the word U^a, all points t^{m-1}/u, with the v-factor at any position $j+1$, where it is simply evaluated; (iii) the mirror image V^a.

String weight. A string of length A with base q contributes $(-1)^{A-1}\varphi_{A-1}(t)\,q^A$: the seed residue is q; a successor pair contributes the residue $qt^\alpha(t-1)$; a non-successor pair at string distance $\delta\geq 2$ contributes $\frac{t^\delta-1}{t^\delta-t}$; and $\prod_{\delta=2}^{A-1}\big(\frac{1-t^\delta}{t(1-t^{\delta-1})}\big)^{A-\delta}=t^{-\binom{A-1}{2}}\varphi_{A-1}/(1-t)^{A-1}$.

Interaction. For a U-string of length A and a V-string of length $B=a-A$, the cross factors are $\frac{wt^{\beta-\alpha}-1}{wt^{\beta-\alpha}-t}$ (U_α before V_β) and $\frac{1-wt^{\beta-\alpha}}{1-wt^{\beta-\alpha+1}}$ (V_β before U_α), and their sum over shuffles is the function $\mathrm{Sh}_{A,B}(w)$ of Lemma 9.4.

Assembling. Since $g(z^*)=u^{-d}G_A(w)$ for configuration (i) and $u^{-d}g(\pi_a)$ for (ii), where $\pi_a=(1,t,\dots,t^{a-1})$, and $v^{-B}=u^{-B}w^B$,

$$\varphi_a\rho(w)=\sum_{A=1}^{a-1}(-1)^a\varphi_{A-1}\varphi_{B-1}\,w^B\,\mathrm{Sh}_{A,B}(w)\,G_A(w)+(-1)^{a-1}\varphi_{a-1}\,g(\pi_a)\sum_{j=0}^{a-1}\frac{w}{t^j-w}+O(w^n),$$

the last term coming from (iii). This is an identity of rational functions of w, each term analytic at $w=0$, so $[w^y]$ may be taken termwise. By Lemma 9.4, $\mathrm{Sh}_{A,B}(w)=\begin{bmatrix}a\\A\end{bmatrix}_t t^{-AB}(1+\kappa_{AB}\sum_{m\geq1}(t^{-Am}-t^{Bm})w^m)$ with $\kappa_{AB}=\frac{(1-t^A)(1-t^B)}{1-t^a}$, and $\begin{bmatrix}a\\A\end{bmatrix}_t\varphi_{A-1}\varphi_{B-1}=\frac{\varphi_a}{(1-t^A)(1-t^B)}$, $\frac{\varphi_{a-1}}{\varphi_a}=\frac{1}{1-t^a}$. For $1\leq y\leq n-1$ this gives the formula. □

Lemma 9.4 (Shuffle identity)**.**

$$\mathrm{Sh}_{A,B}(w)=\begin{bmatrix}A+B\\A\end{bmatrix}_t t^{-AB}\,\frac{(1-w)(1-t^{B-A}w)}{(1-t^{-A}w)(1-t^Bw)},$$

$$\frac{(1-w)(1-t^{B-A}w)}{(1-t^{-A}w)(1-t^Bw)}=1+\kappa_{AB}\sum_{m\geq1}(t^{-Am}-t^{Bm})w^m.$$

Proof. Induction on $A+B$; for $A=0$ or $B=0$ both sides are 1. Condition on the last letter. If it is U_{A-1}, its cross factors telescope to $\frac{1-wt^{1-A}}{1-wt^{B+1-A}}$; if it is V_{B-1}, to $t^{-A}\frac{1-wt^{B-1}}{1-wt^{B-1-A}}$. So $\mathrm{Sh}_{A,B}=\frac{1-wt^{1-A}}{1-wt^{B+1-A}}\mathrm{Sh}_{A-1,B}+t^{-A}\frac{1-wt^{B-1}}{1-wt^{B-1-A}}\mathrm{Sh}_{A,B-1}$. With $X=t^A$, $Y=t^B$, $\begin{bmatrix}A+B-1\\A-1\end{bmatrix}_t/\begin{bmatrix}A+B\\A\end{bmatrix}_t=\frac{1-X}{1-XY}$ and $\begin{bmatrix}A+B-1\\A\end{bmatrix}_t/\begin{bmatrix}A+B\\A\end{bmatrix}_t=\frac{1-Y}{1-XY}$, the claimed closed form satisfies this recursion if and only if $Y(1-X)(1-w/X)+(1-Y)(1-Yw)=(1-XY)(1-wY/X)$, and both sides equal $1-XY-wY/X+wY^2$. For the expansion: the left side is 1 at $w=0$, tends to 1 at ∞, and has simple poles with residue $-t^A\kappa_{AB}$ at $w=t^A$ and $t^{-B}\kappa_{AB}$ at $w=t^{-B}$; so does the right side. □

Remark 9.5 (Relation to Green polynomials). Write $p_\mu=\sum_\lambda\mathcal{X}^\lambda_\mu(t)P_\lambda(x;t)$. For $g=P_\rho$ and $\lambda=\rho+1^a$ one has $\Phi_a(P_\rho;x,y)=(1-t^x)(1-t^y)\mathcal{X}^\lambda_{(x,y)}(t)/b_\lambda(t)$, so Theorem 9.3 evaluates Green polynomials at two-part classes; the two t-strings are the Frobenius orbits of a maximal torus of type (x,y) [Gre55]. Theorem 9.2 is, up to the normalising factors $b_\lambda(t)$ and $\prod_i(1-t^{\mu_i})$, Jing's constant-term expression for the matrix element $\langle Q_\lambda,p_\mu\rangle$ [Jin91], cf. [JL21, (2.17), (2.20)]. Jing and Liu [JL21, Thm. 2.7] expand the same matrix element as a nested sum valid for all μ, extracting one variable at a time; at two-part μ that sum does not close up, since its inner sums run over classes of arbitrary length. Their closed forms [JL21, Thm. 2.10, (2.37)–(2.40)] restrict the Hall–Littlewood index λ, not the class μ; their Murnaghan–Nakayama rule [JL21, Thm. 3.2] generalises a formula of Morris [Mor77][1], its case $\ell(\lambda)=2$. We evaluate by residues instead, and at $\ell(\mu)=2$ the poles organise into two t-strings. The formula is explicit, with no sum over

[1]We have not seen [Mor77] first-hand; its scope is as described in [JL21, p. 12, before Thm. 3.2].

classes, and linear in the two-string specialisation G_A of g; for general ρ, G_A is itself a finite (branching) sum, so Theorem 9.3 is not a product formula.

Example 9.6 (Two rows)**.** For $a=2$, only $A=B=1$ occurs, and $G_1(w)=P_\rho(1,w;t)=w^{\rho_2}\frac{(1-tw)-w^r(w-t)}{1-w}$ with $r=\rho_1-\rho_2\geq 1$ ($G_1=w^{\rho_2}$ if $r=0$). Substituting into Theorem 9.3 gives, for $\lambda=(\lambda_1,\lambda_2)$, $y\leq x$, and $m_{xy}=2$ if $x=y$, 1 otherwise,

$$\mathcal{X}^{(\lambda_1,\lambda_2)}_{(x,y)}(t)=\begin{cases}(t-1)\,t^{\lambda_2-1-y}(1+t^y), & y<\lambda_2,\\ m_{xy}-(1-t)\,t^{\lambda_2-1}, & y=\lambda_2,\\ (t-1)\,t^{\lambda_2-1}, & y>\lambda_2,\end{cases}$$

in agreement with a Kostka–Foulkes computation for $|\lambda|\leq 10$. This is not new: for $\lambda=(n-k,k)$ and every class ρ, Jing and Liu [JL21, Thm. 2.6, unnumbered display after (2.32), same in v1 and v2] give $\mathcal{X}^\lambda_\rho=\pi_k+(t-1)\sum_{j<k}t^{k-1-j}\pi_j$, with π_j the number of j-subsets fixed by ρ (their $D^{(j)}(\rho)$). A short proof: $\mathcal{X}^\lambda_\rho=\sum_\nu\chi^\nu_\rho K_{\nu\lambda}(t)$ [Mac95, III (7.6′)], where $K_{(n-j,j),\lambda}=t^{k-j}$ (one tableau, and K is monic of degree $n(\lambda)-n(\nu)$ [Mac95, III (6.5)]) and $\chi^{(n-j,j)}_\rho=\pi_j-\pi_{j-1}$ by Young's rule. On the diagonal class $(x,y)=\lambda$ it gives $t^{\lambda_2}-t^{\lambda_2-1}+m_{xy}$, which for $\lambda_1>\lambda_2$ with $\lambda_2\geq 3$ odd equals -1 at $t=-1$ and 1 at $t=0$. A root in $(-1,0)$ rules out any product or quotient of cyclotomic factors and powers of t, so Green polynomials at two-part classes admit no product formula even for two rows; for expressions $t^c\prod_i(1-t^{d_i})^{e_i}$ with $e_i\in\mathbb{Z}$ (products and quotients) this was observed independently in [Cli26, Thm. D].

Theorem 9.7 (Closed $\ell=3$, $\kappa=1$ leads)**.** *Let $\lambda=(a,b,c)$ be any ordering of a length-3 partition, $\mu=(x,y)$ with $\kappa(\lambda,\mu)=1$, $n=x+y$, $m_{xy}=2$ if $x=y$ and 1 otherwise. Put*

$$\Xi_a(r,q):=\operatorname{lin}_e T_a(p_rp_q)=(-1)^{r+q}\frac{[a+r+q]}{[a]}[a]_{t^r}[a]_{t^q}$$

and $U_a(b,c;x,y):=\big((-1)^n\Phi_a(p_bp_c;x,y)-\Xi_a(b,c)\big)/m_{xy}$. Then

$$[(s-1)^2]\,c_{\lambda\mu}=(-1)^{b+c}U_a(b,c;x,y)+\sum_{\substack{1\leq r<b\\ \{b-r,\,a+r+c\}=\{x,y\}}}(-1)^{r+c}\Xi_a(r,c)+\sum_{\substack{1\leq q<c\\ \{c-q,\,a+b+q\}=\{x,y\}}}(-1)^{b+q}\Xi_a(b,q)+[e_xe_y]\,D_a(M_{bc}),$$

with D_a the derivation $D_a(e_j)=M_{aj}$ of Proposition 6.2.

Lemma 9.8. *For G homogeneous of degree $n=x+y$, $\langle G,p_xp_y\rangle=(-1)^n\big(m_{xy}[e_xe_y]G+\operatorname{lin}_e G\big)$.*

Proof. $\langle e_\nu,p_xp_y\rangle=\langle\Delta e_\nu,p_x\otimes p_y\rangle$ since p_x,p_y are primitive, and $\langle e_m,p_m\rangle=(-1)^{m-1}$. Each tensor factor must be a single e, so $\ell(\nu)\geq 3$ gives 0, $\langle e_n,p_xp_y\rangle=(-1)^n$, $\langle e_xe_y,p_xp_y\rangle=(-1)^nm_{xy}$, and other two-part ν give 0. □

Proof of Theorem 9.7. By Proposition 9.1, $[(s-1)^2]c_{\lambda\mu}=[e_xe_y]\big(\Gamma_a(e_b,e_c)+D_a(M_{bc})\big)$. In $\Gamma_a(e_b,e_c)=\sum_{r,q}(-1)^{r+q}e_{b-r}e_{c-q}T_a(p_rp_q)$, terms with $r<b$ and $q<c$ have at least three factors in I and do not contribute. For $r<b$, $q=c$, only the single-e part $\Xi_a(r,c)e_{a+r+c}$ of $T_a(p_rp_c)$ (Lemma 6.11) contributes; symmetrically for $r=b$, $q<c$. For $r=b$, $q=c$ the contribution is $(-1)^{b+c}[e_xe_y]T_a(p_bp_c)=(-1)^{b+c}U_a(b,c;x,y)$ by Lemma 9.8 and $\operatorname{lin}_e T_a(p_bp_c)=\Xi_a(b,c)$. □

Corollary 9.9. *Every lead at valuation 2 ($\kappa=\ell-2$, any ℓ) has a closed formula: by Theorem 7.8 it is a sum of products of pair leads (Proposition 6.2) and $\ell=3$, $\kappa=1$ leads (Theorem 7.4, Corollary 8.3, Theorem 9.7).*

Example 9.10. $(3,3,3) \to (7,2)$: Lead $= 2t^{13} + 3t^{12} + 3t^{11} + 6t^{10} + 6t^9 + 6t^8 + 9t^7 + 6t^6 + 3t^5 + 9t^4 + 6t^3 + 3t + 4$ (agrees with direct computation). Leads are *not* positive in general: $(4,4,2) \to (7,3)$ has Lead $= (t+1)(t^2+1)(2t^8 + t^7 + t^6 + 3t^4 + t^3 + t^2 - t + 2)$.

Computer checks. All statements in this section were checked against an independent implementation of Hikita's subset formula: Theorem 9.7 on 16 pairs with $n \le 12$ (both examples among them) and on 27 pairs in all 3 orderings with $n \le 10$; Theorem 9.3 in 229 cases ($a \le 5$); and the Green-polynomial dictionary symbolically for $|\lambda| \le 8$ and numerically for $|\lambda| \le 9$.

10. Open problems

(1) **Valuation ≥ 3.** Connected leads with $\ell \ge 4$, $\kappa = 1$. Hierarchy: $v = 1 \leftrightarrow p_n \leftrightarrow$ one t-string (Lemma 6.11); $v = 2 \leftrightarrow p_x p_y \leftrightarrow$ two strings (Theorem 9.3). Does the three-string shuffle sum factor pairwise?

(2) **Zero set at generic t.** At $t = -2$ seven pairs with $n = 6$ have $v > \ell - \kappa$; for coarsenings this is cancellation between histories, e.g. $\mathrm{Lead}_{(1,1,1),(3)} = [3]_t(2+t)$. Describe the zero set of $[(s-1)^{\ell-\kappa}]c_{\lambda\mu}$ in t.

(3) **Counting.** A formula for $N(\lambda,\mu)$ in Theorem 6.6; for $\ell(\lambda) = 3$ and $\mu = (n-1,1)$ with $\kappa = 1$ we observe $\mathrm{Lead}_{\lambda\mu}(1) = 2n^2 - 6n + 3$, independent of λ ($n \le 10$; computed only).

(4) **Positivity.** Leads are not t-positive (Example 9.10), so no "t-count of minimal flows" can hold. Is there a signed or cancellation-free model?

(5) **Cumulant link.** Is there a Macdonald-cumulant construction whose leads are the $\star$-leads for all t, i.e. cumulants of $e_k \mapsto t^{\binom{k}{2}}$?

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